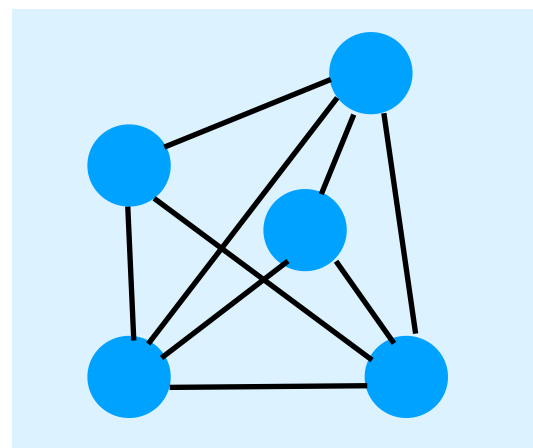


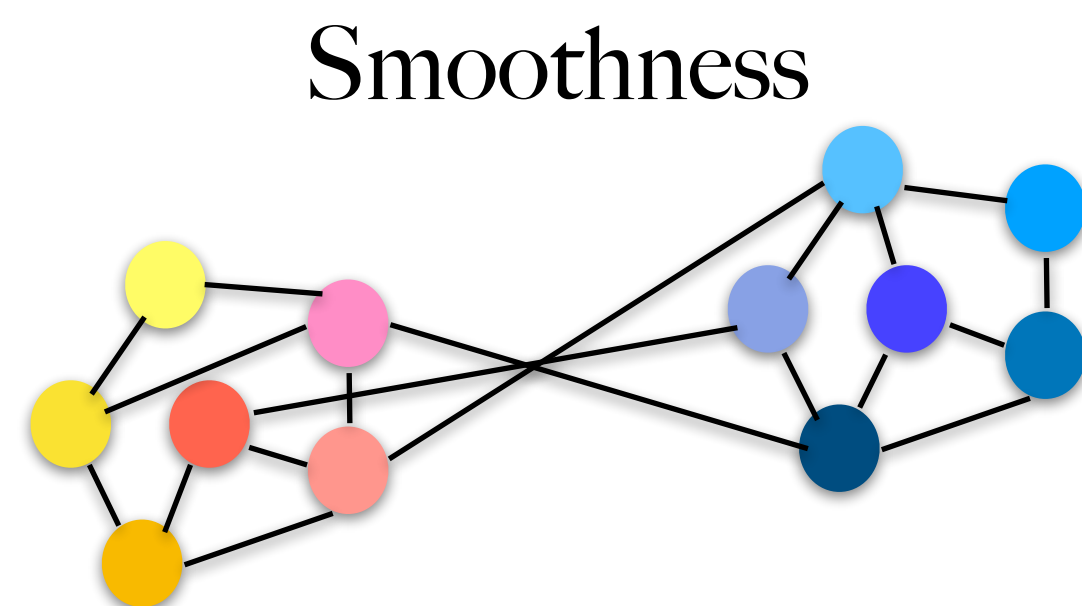
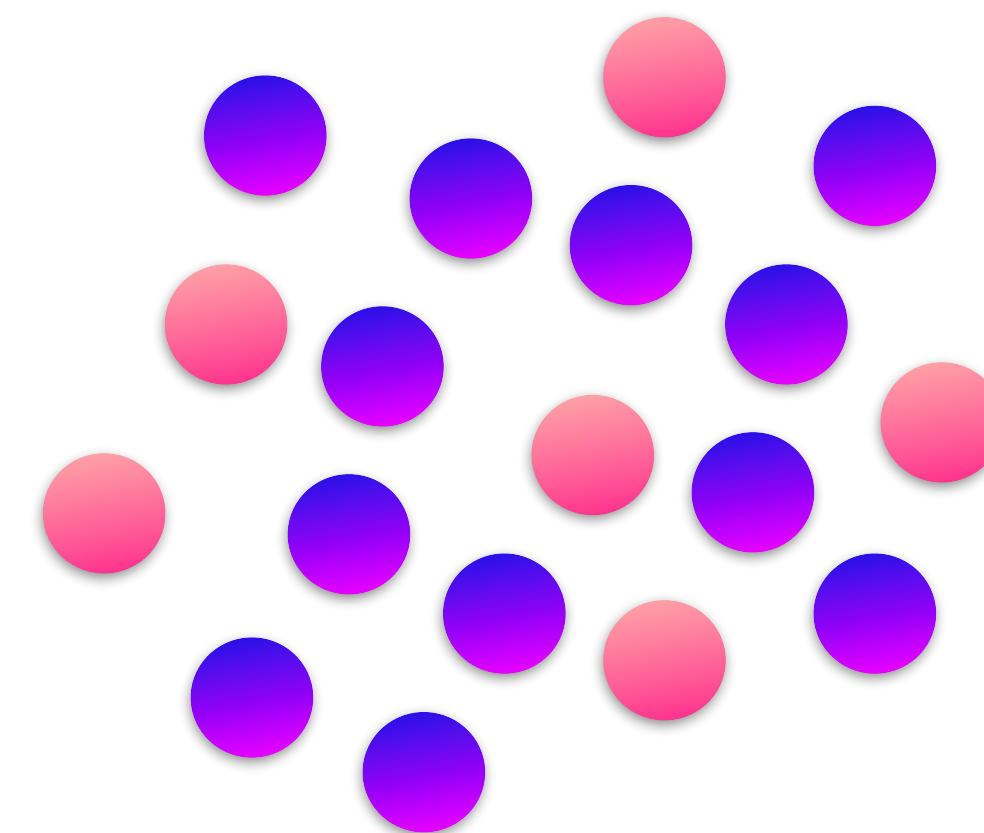
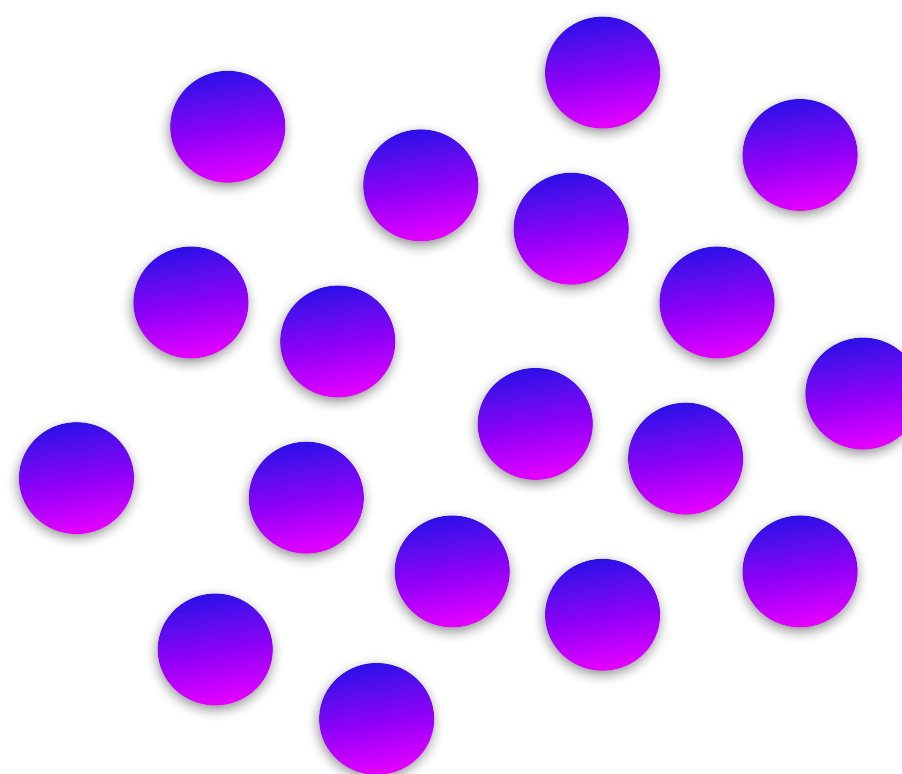
Graph Learning & Node Subset Selection Under Sparsity Constraint



	T1	T2	T3	T4	T5	T6
1	■	■	■	■	■	■
2	■	■	■	■	■	■
3	■	■	■	■	■	■
4	■	■	■	■	■	■

Graph Learning

Node Subset Selection

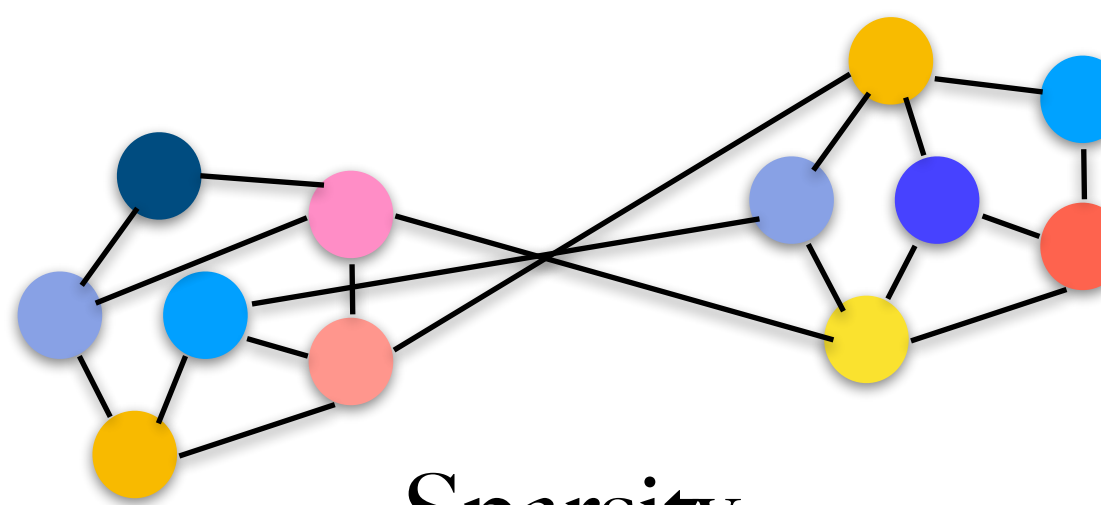


Smoothness

If Data



Says That



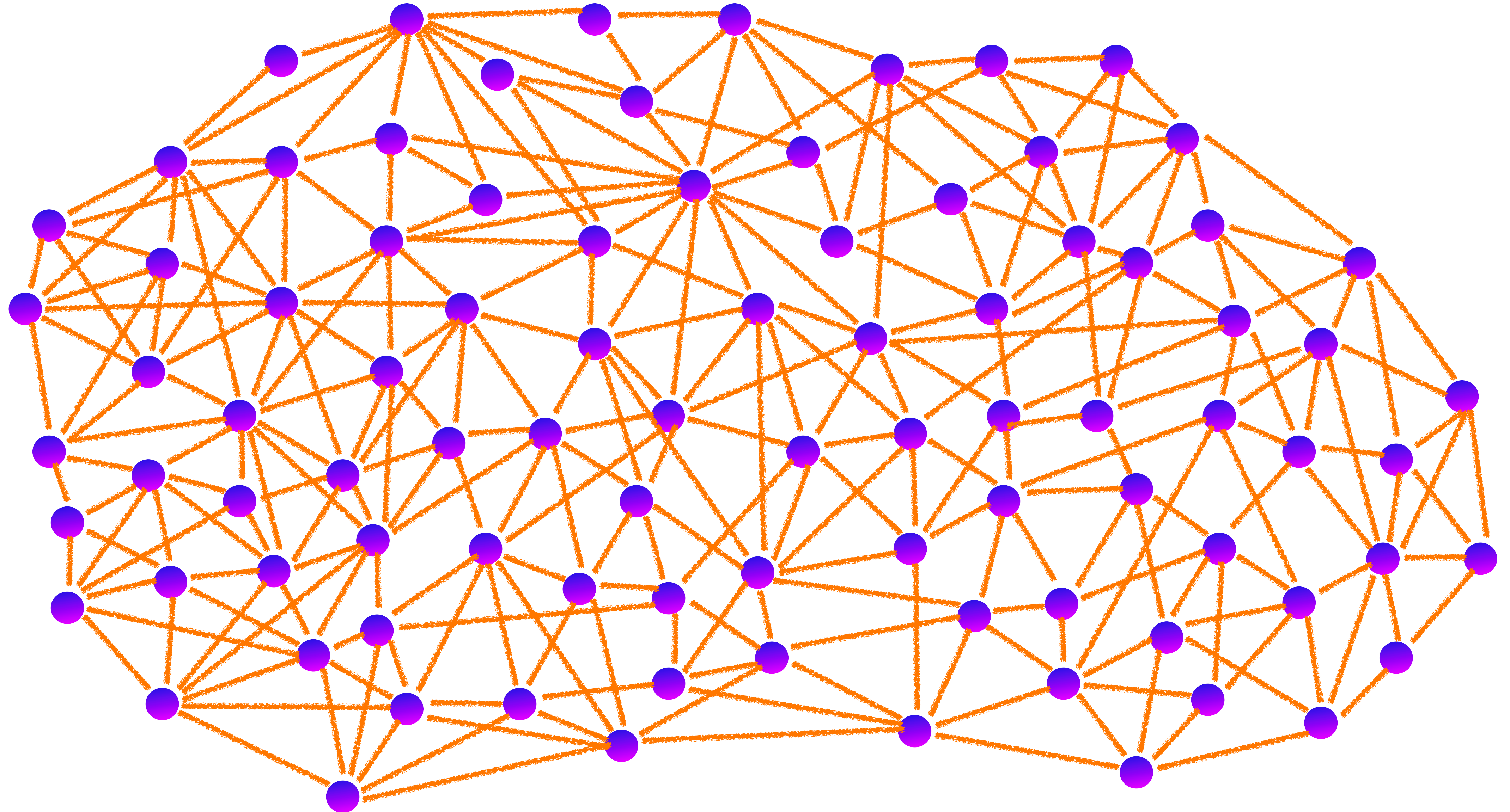
Sparsity

Under Sparsity Constraint

Lets talk about Subset Selection

To Motivate

Lets Go Back Back in the Days of Novel Corona Virus Pandemic



1ST

2ND

3RD

4TH

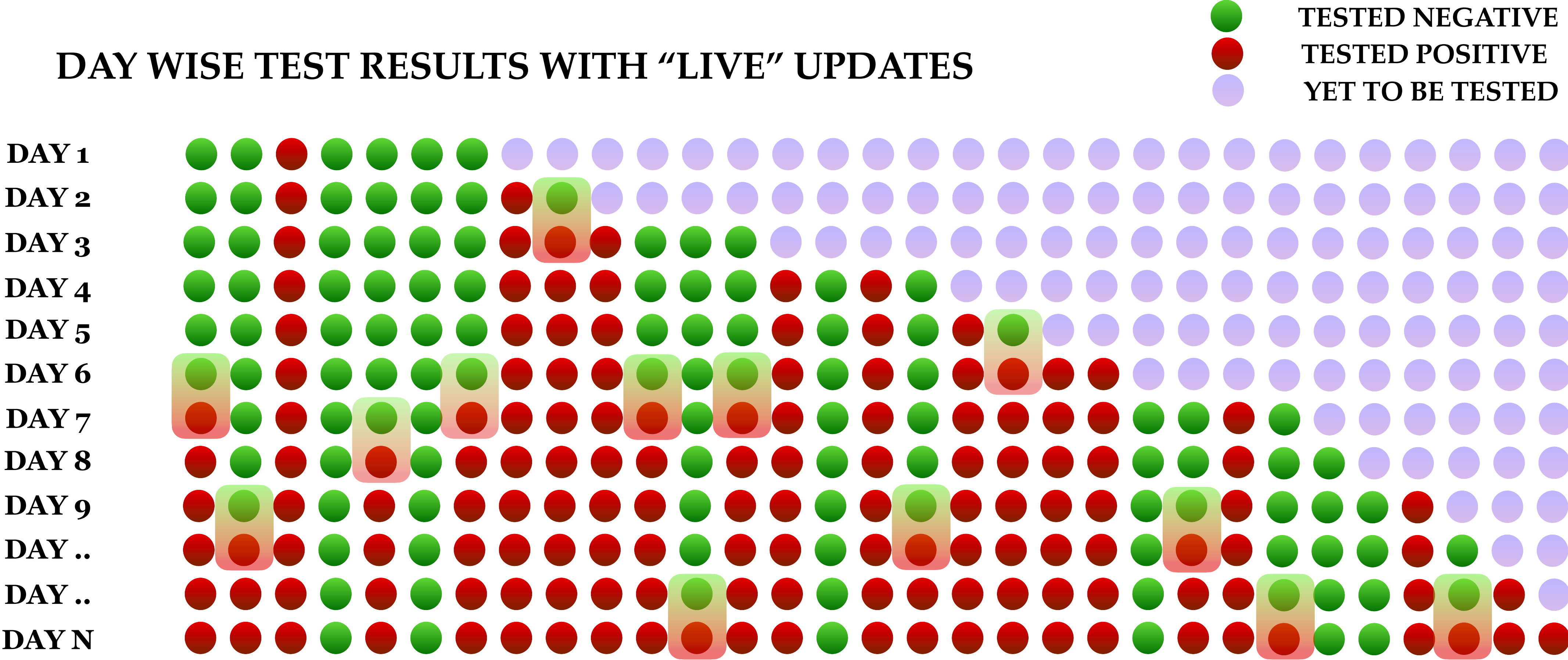
5TH

6TH

7TH



Testing Paradox



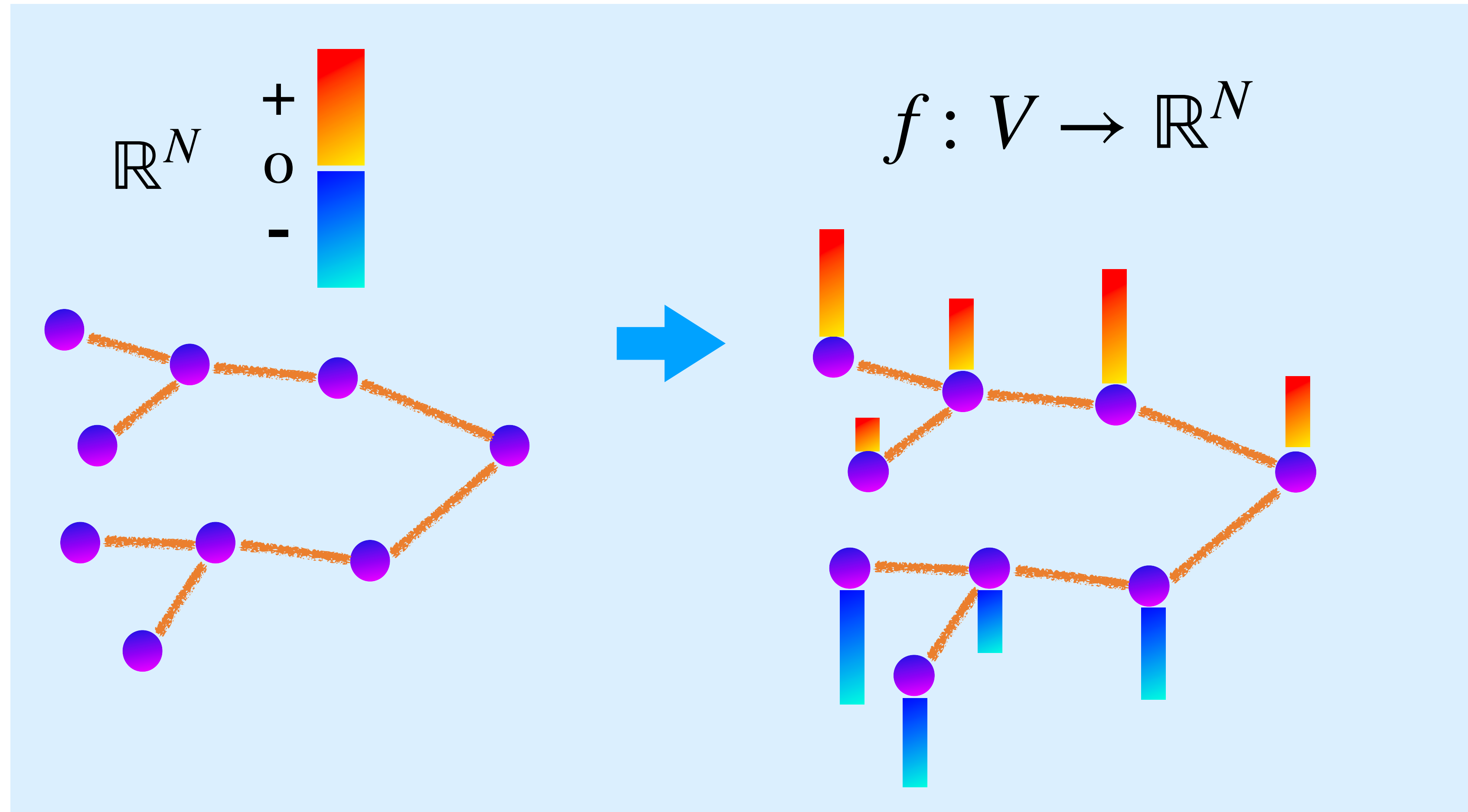
NAIVE TESTING PROCEDURE APPLIED IN EPIDEMIC

GSP

Graph Signal Processing

GSP 101

Graph Signal Processing

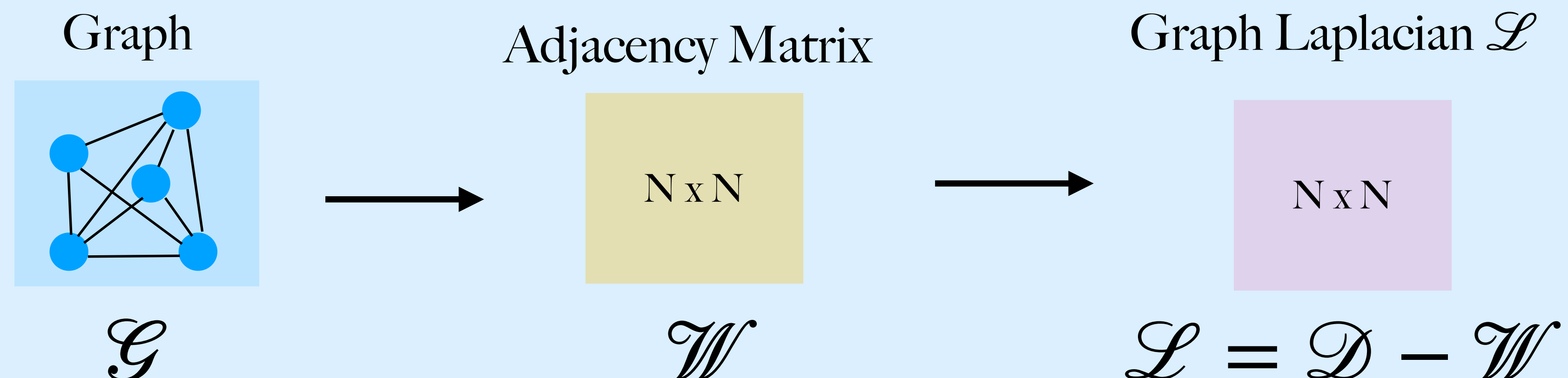


Graph
Representation

Graph Signal Processing

But we wanted to apply Signal
Processing on Graphs! :(

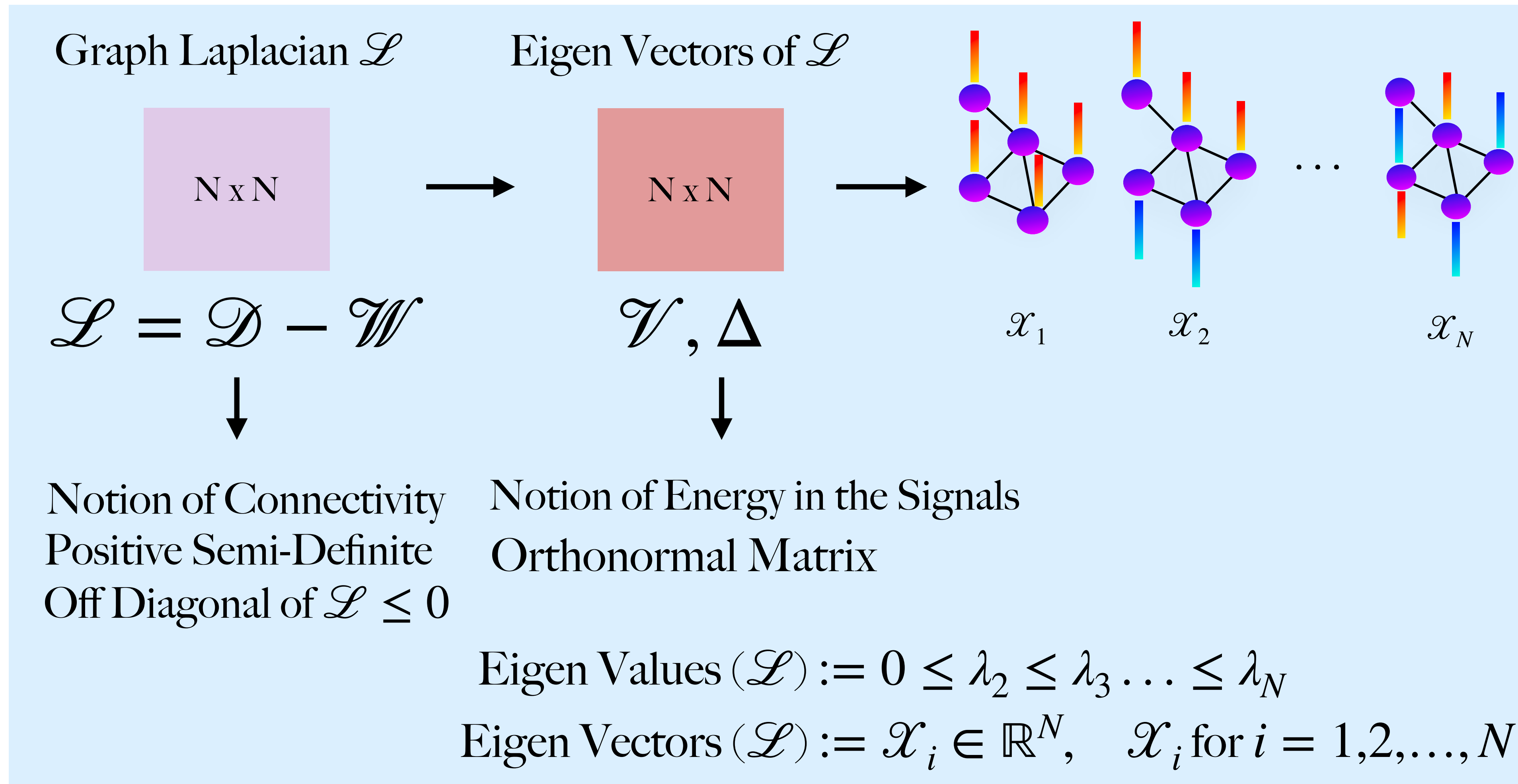
There is some representation of
Graphs that can help us in that...



Signal
Processing

Still Where is
Frequency Notion ?

Graph Signal Processing



Signal
Processing

CS

Compressed Sensing

Let $X \in \mathbb{R}^M$ and $Y \subseteq X$
 $\text{size}(Y) = K$

C be a Subset Selector Matrix

$$Y = C \cdot X$$

Based on the subset Y , we want to reconstruct X

Sparse Signal Reconstruction

Optimise for $X \quad \|Y - C \cdot X\|_2^2$

But this has **Infinitely** Many Solutions

Background of
 Problem that
 CS Solves

So how could we obtain X uniquely ?

How can CS help in this ?

CS takes the advantage of the fact that many signals are **Sparse** i.e. they contain many coefficients close to or equal to zero, when represented in **some Domain**.



Attempt to
Perform
CS

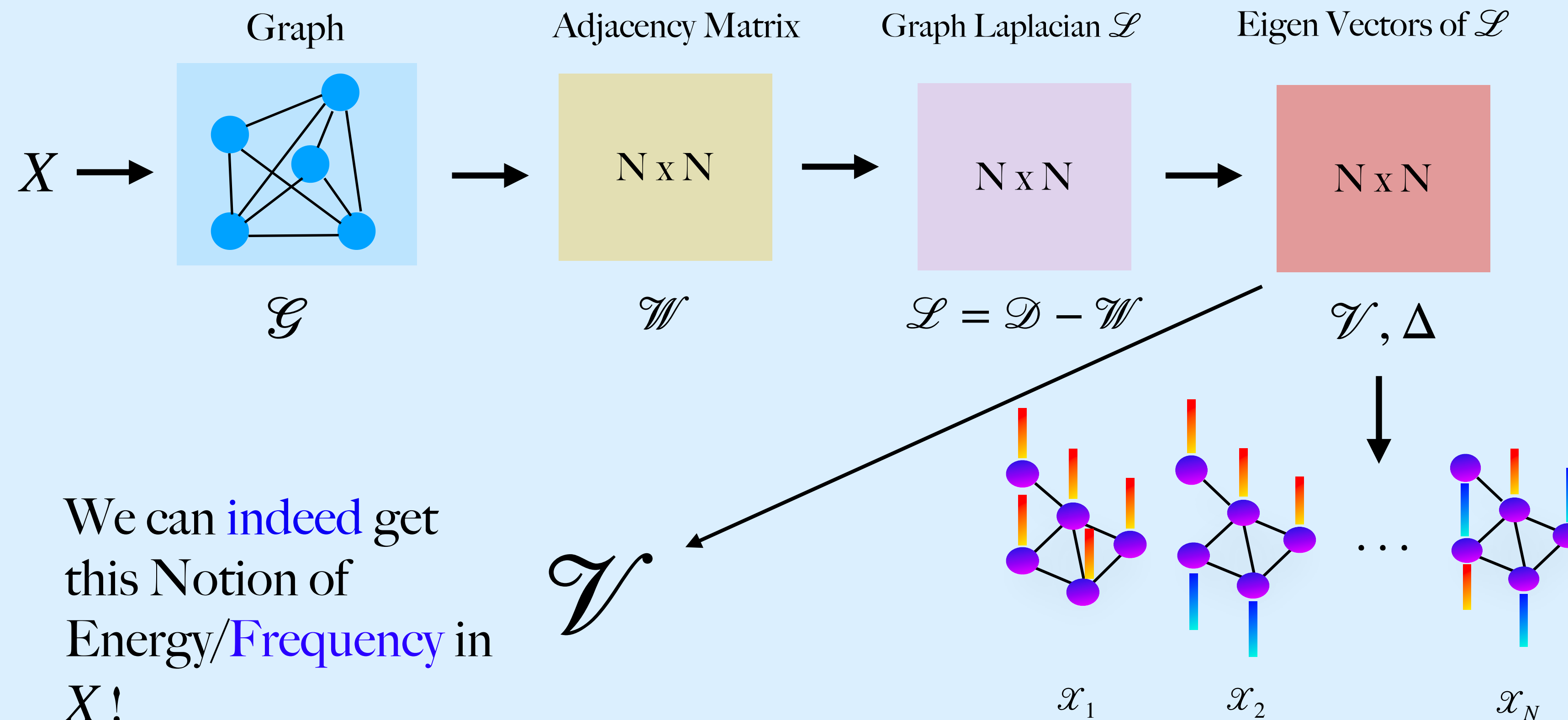
$$X \cdot \psi = s$$

Law of Conservation of Energy $\implies$ Total Energy in this System should be Preserved!

$$X = \psi^{-1} \cdot s$$

$\implies$ Components of ψ^{-1} must Capture **Notion of Energy/Frequency**

Now from where we can get this Notion of Energy/Frequency in X ?



Attempt to
Perform
CS

$$\text{So } \psi^{-1} = \mathcal{V} \implies X = \mathcal{V} \cdot s$$

Recall Subset Selection Problem that we began with

$$[U] = [V] \begin{bmatrix} \text{Grid with 6 blue squares} \end{bmatrix} [V^T] [S]$$

$$Y = C \cdot X \implies C \cdot \mathcal{V} \cdot s$$

Attempt to Perform CS

But CS says something more . . .

CS says, Not all Under-Determined System of Linear Equations have a Sparse Solution.
However if there exists a Unique Sparse Solution to the Under-Determined System,
then the Compressed Sensing Framework allows the recovery of that Solution.

So, $X = \mathcal{V} \cdot s$ i.e. Exact Recovery Only possible when s is the sparsest

So for Nice Recovery We propose the Following

$$s^* = \min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

Recovery

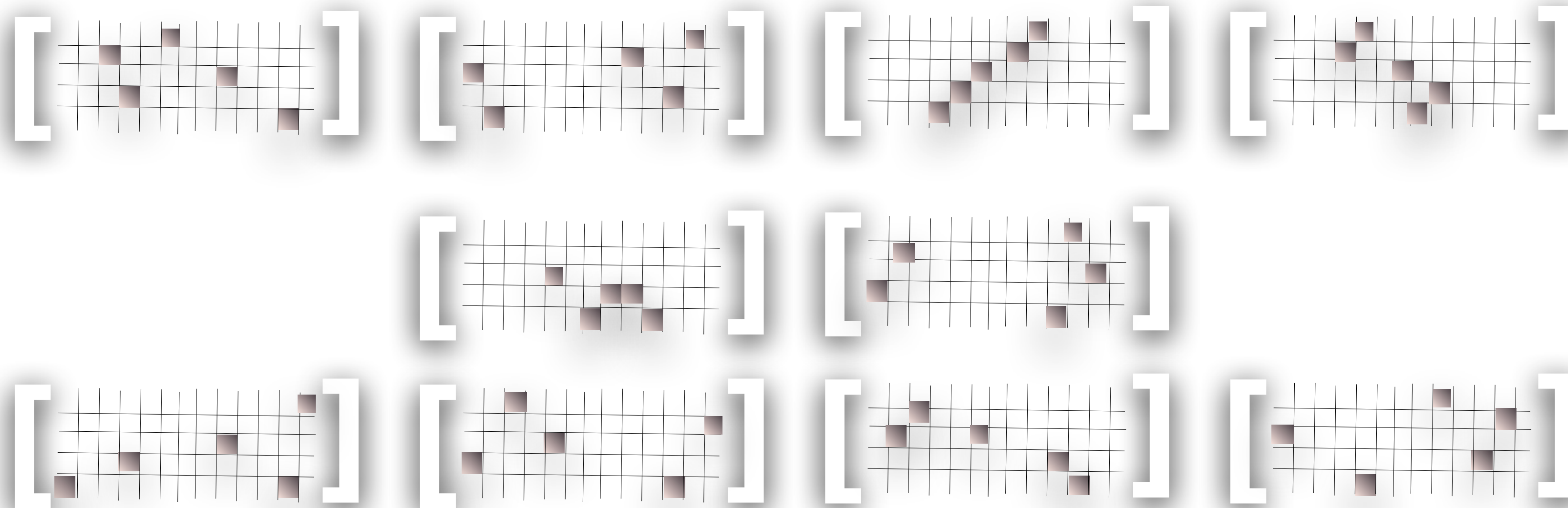
$$X^* = \mathcal{V} \cdot s^*$$

Attempt to
Perform
CS

CS 101

Performance of this Model **depends** upon the Choice of Subset, so
Subset Selection Matrix Needs a Honourable Mention.

Subset Selector Matrix *C* 101



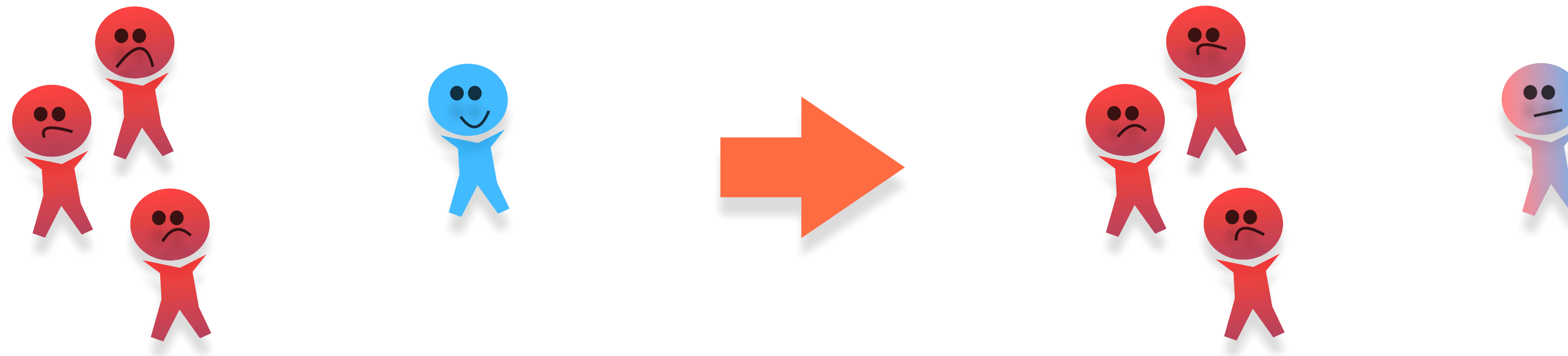
Attempt to
Perform
Subset
Selection

So there are Variety of Way to Select a Subset

Let's Try To Perform **Subset Selection** to Efficiently Handle
Measles Outbreak in Mumbai

Step 0. What is Measles ?

A DISEASE WHICH SPREADS EASILY FROM PERSON TO PERSON THROUGH AIR, AND CAN CAUSE FEVER, COUGH, RASH.



Step 1. Analyse the Data Set

Example Setting Considers Dataset from Measles dataset from BMC

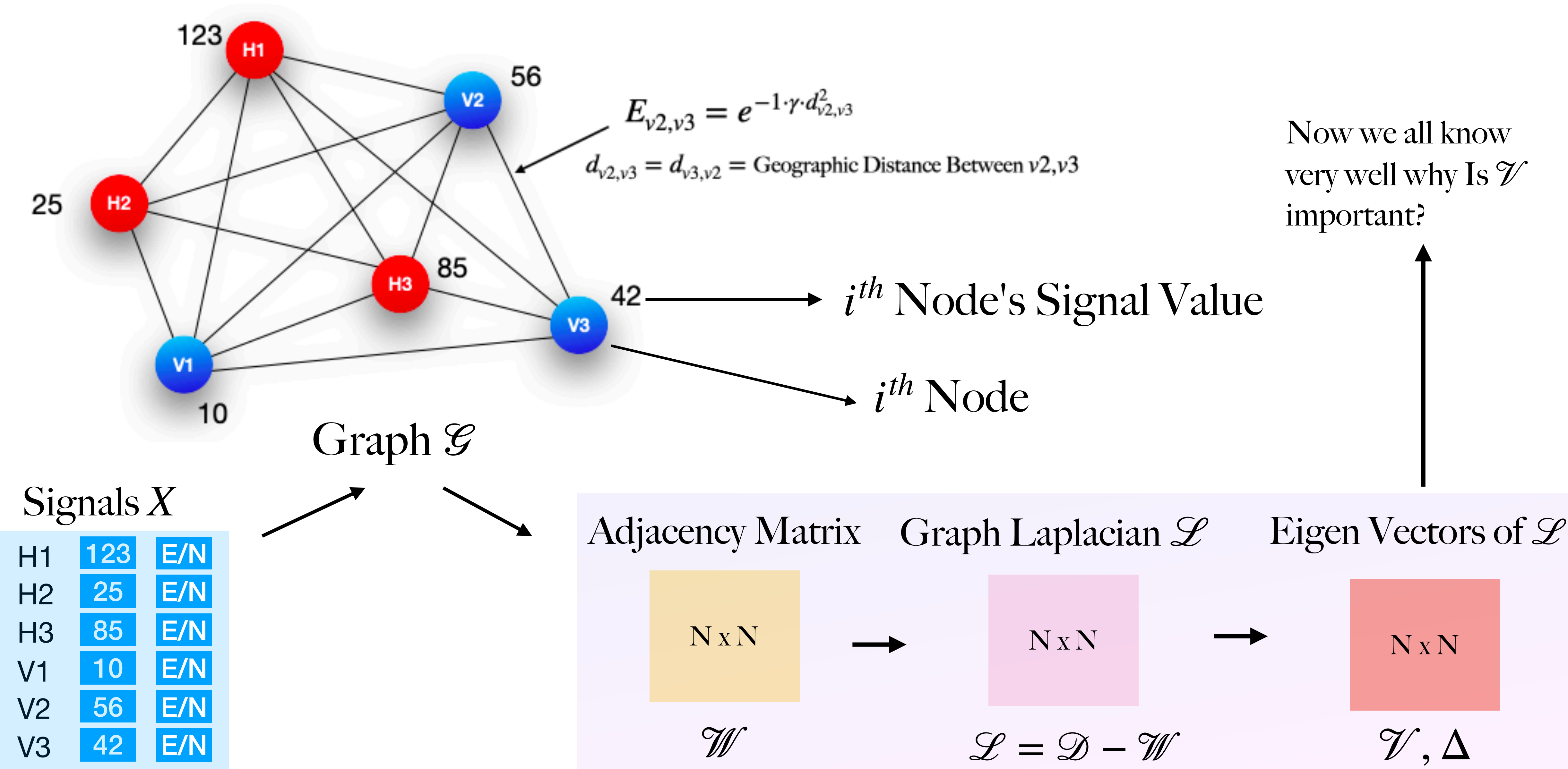
Name of Patient	Address/Village	Hospital Registered With	Ward	Measles Y/N
Person 1	Colaba	JJ Hospital	A	Y
Person 2	Antop Hill	Cooper Hospital	B	N
Person 3	Govandi	Kasturba Hospital	M	Y
Person 4	Andheri	Lilavati Hospital	L	Y

Step 2. We Calculated How many cases were Registered From Village or at a Hospital.

Hospital	# Cases	Lat/Long	Village	# Cases	Lat/Long
 H1	123	19'E, 24N	 V1	20	19'E, 24N
 H2	45	19'E, 24N	 V2	125	19'E, 24N
 H3	13	19'E, 24N	 V3	25	19'E, 24N

Every Entity in Population Related with a Scalar Signal Value i.e. #Cases
Let there be N Entities in Total. $X \in \mathbb{R}^N$, is a Vector of Health Stats of Entities
Entities $\in \{H, V\}$

Step 3. GSP Interpretation of Setting Developed in Last Step



Step 4. Compressed Sensing Model for Optimal Reconstruction

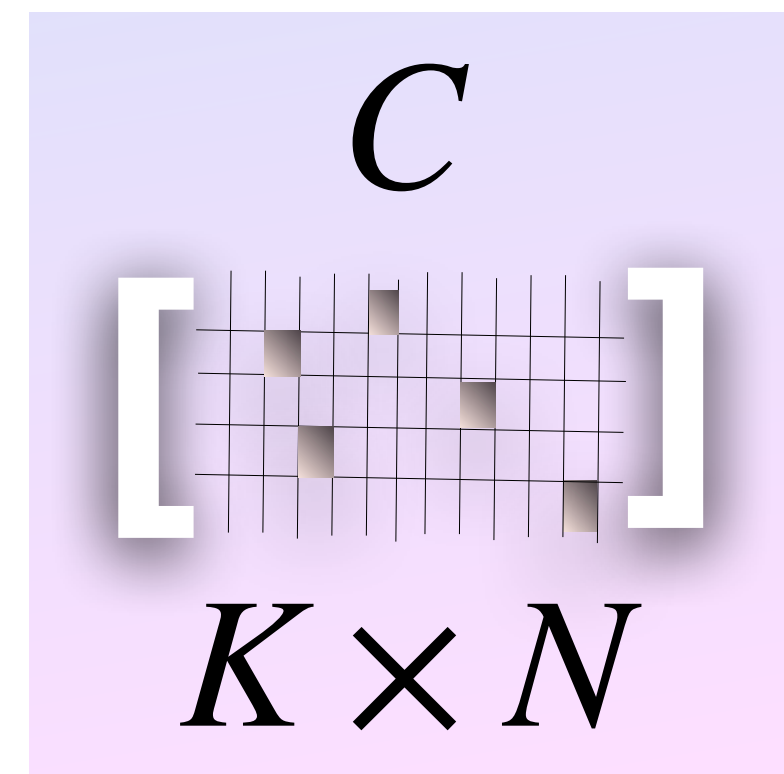
For obtaining a **Reconstruction** of the Health Status over the **Entire Population** we need to **Solve** the Following, and obtain the s

$$\min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

Select an Appropriate Subset Selector Matrix C

$$\min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

✓ ✓ ✓ ?



Now we can solve the above optimisation problem, to obtain s^*

*Hyperparameter λ is obtained via Cross-Validation

Step 5. Perform Reconstruction of Stats across Population

$$X^* = \mathcal{V} \cdot S^*$$

Reconstruction Obtained

NEXT

Selection of an Appropriate Subset Selector Matrix C

We go over 3 Heuristics viz. Node-Degree, Kemeny and Mutual Coherence

Subset Selector Matrix 101

1. Node-Degree

Algorithm 1 Node Degree based Subset Selection Algorithm

Initialize $node_degrees$ as a vector of zeros with length N .

for $i = 1$ to N **do**

$node_degrees(i) \leftarrow D(i, i)$

end for

Sort the $node_degrees$ vector in descending order, preserving the index order.

Let m be the desired subset size, e.g., $m = 500$.

Create an array idx_mix containing the first m elements from the sorted index vector.

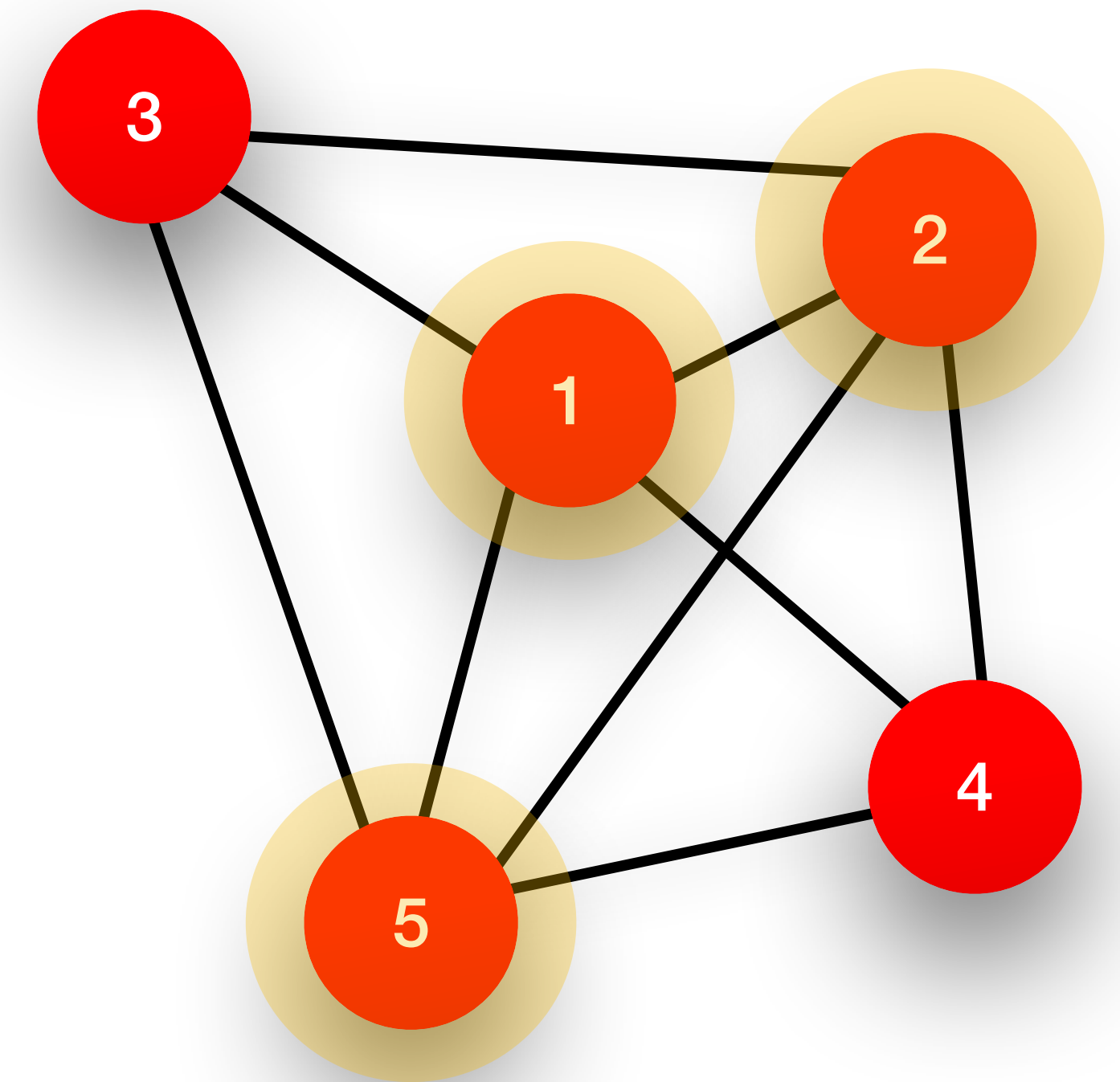
Initialize a vector y with m elements, and a matrix C with m rows and N columns, all initialized to zeros.

for $i = 1$ to m **do**

$y(i) \leftarrow X(idx_mix(i))$

$C(i, idx_mix(i)) \leftarrow 1$

end for



Node $\in \{1, 2, 5\}$ Selected
for Call of Top-3 Nodes

Subset Selector Matrix 101

2. Kemeny Technique

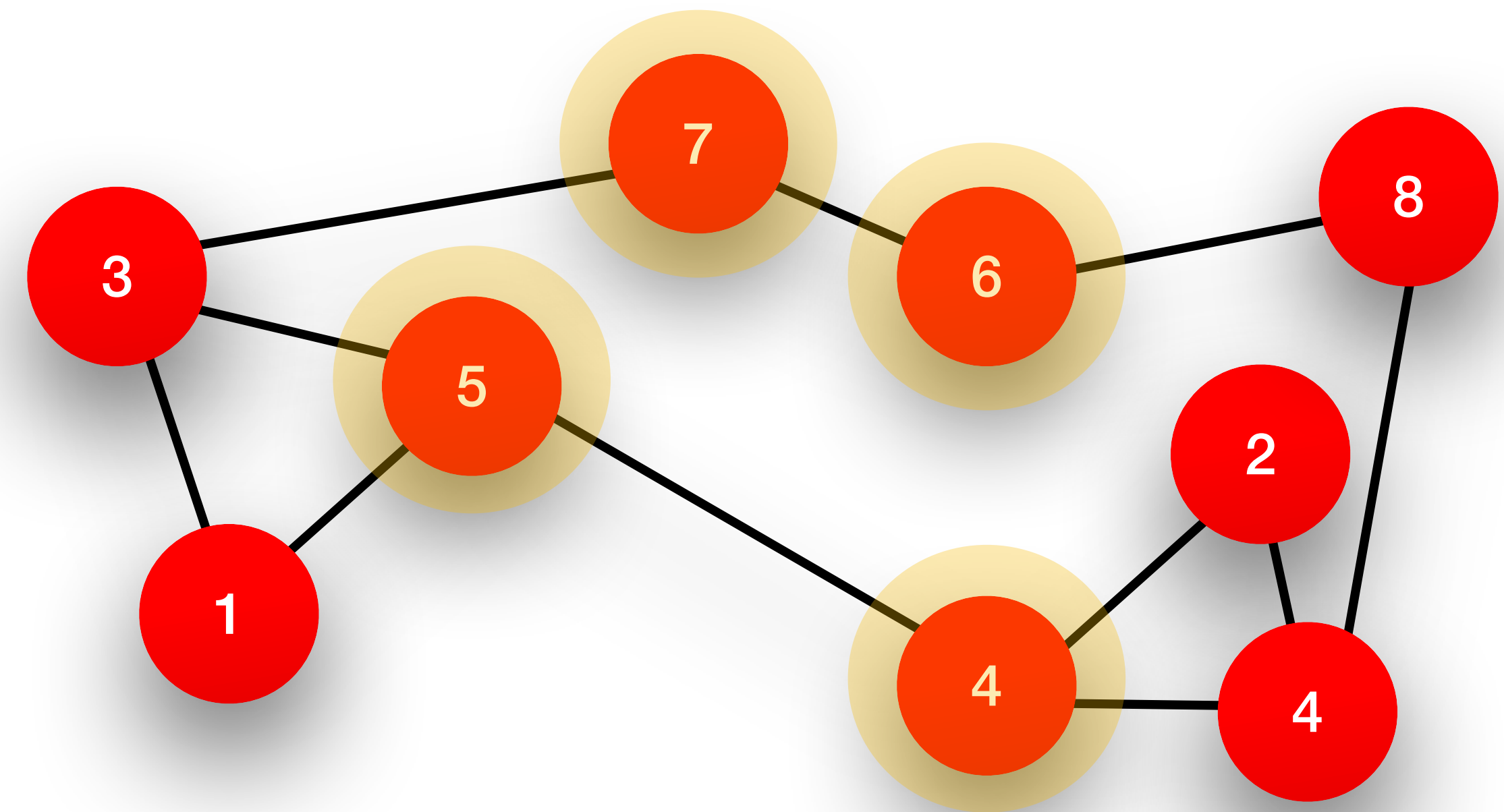
$$K = \sum_{j=1}^n m_{ij} \pi_j$$

Idea Similar to Page Rank $\implies \pi = P \cdot \pi$
 $m_{i,j} = \mathbb{E}[\text{\#Steps } S_i \rightarrow S_j]$
 $\pi_j = \text{Stationary Distribution Probability of being in } S_j$

Algorithm 1 Kemeny-Based Technique

```
1: Input: Adjacency Matrix  $W$  of size  $N \times N$ ,  $K$  vector of zeros of length  $N$ 
2: for  $i = 1$  to  $N$  do
3:   Step 1: Set Skip Node  $i$ 
4:   Initialize  $A_{skip}$  as an  $(N - 1) \times (N - 1)$  matrix from  $W_{-i}$ 
5:   Step 2: Normalize Rows of  $A_{skip}$  to Probabilities
6:   Step 3: Compute Eigenvalues of  $A_{skip}$ 
7:   Step 4: Compute Kemeny Indicator for Skip Node  $i$ 
8:    $K(i) = 0.0$ 
9:   for  $j = 1$  to  $N - 1$  do
10:     $K(i) = K(i) + \frac{1}{1 - \lambda_j}$ 
11:   end for
12: end for
13: Sort the  $K$  vector in descending order, preserving the index order.
14: Let  $m$  be the desired subset size, e.g.,  $m = 500$ .
15: Create an array  $idx\_mix$  containing the first  $m$  elements from the sorted index vector.
16: Initialize a vector  $y$  with  $m$  elements, and a matrix  $C$  with  $m$  rows and  $N$  columns, all initialized to zeros.
17: for  $i = 1$  to  $m$  do
18:    $y(i) \leftarrow X(idx\_mix(i))$ 
19:    $C(i, idx\_mix(i)) \leftarrow 1$ 
20: end for
```

The rationale of this choice is that **nodes** connecting **two communities** are associated with **very high Kemeny indicators**, as **walking** times become much **larger** if such **nodes are removed**.



Node $\in \{4, 5, 6, 7\}$ Selected for Call of Top-4 Nodes, as these have Largest K-Value

Subset Selector Matrix 101

3. Mutual Coherence

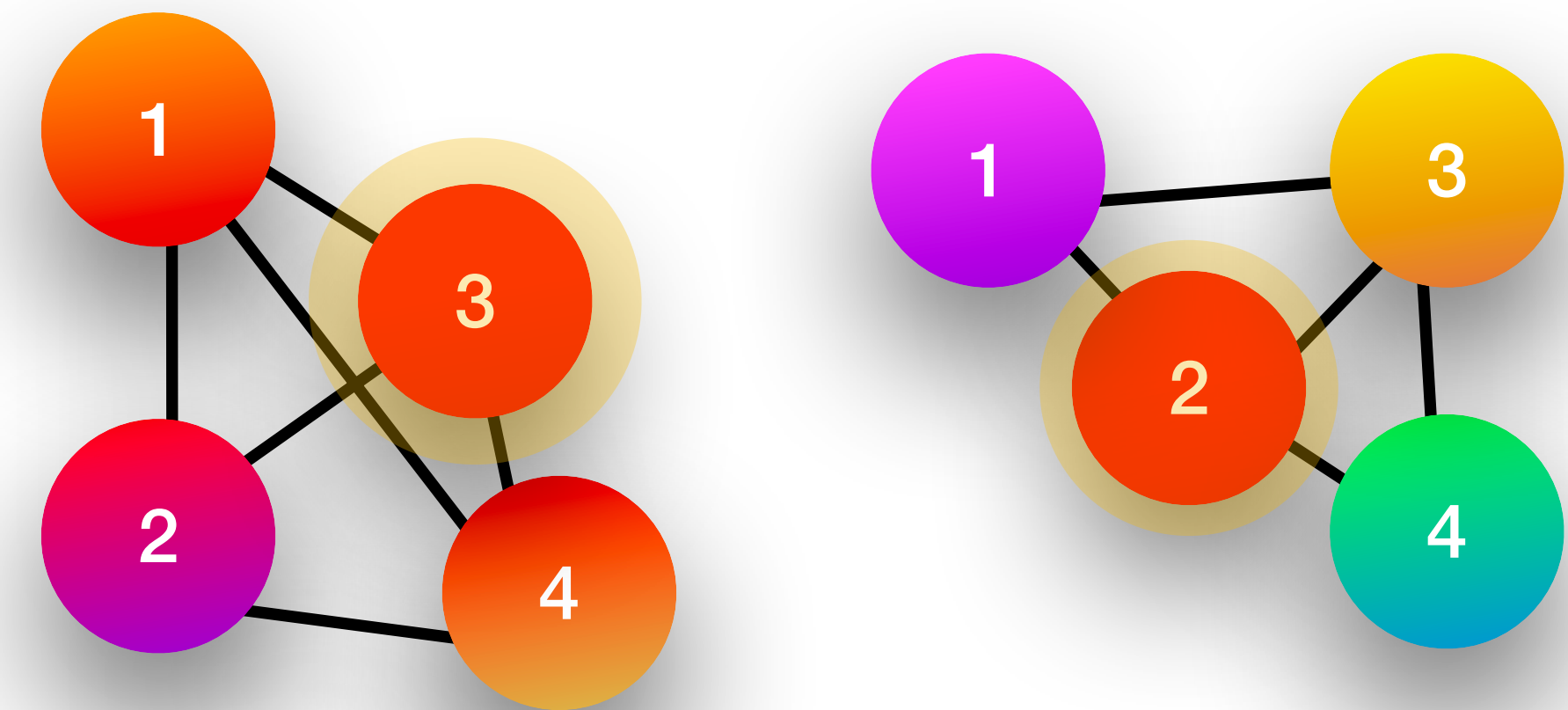
$$\mu_i = \max_j \left(\left| (C \cdot V)_{ij} \right| \right) \quad \text{for } i = 1, 2, \dots, M$$

Algorithm 1 Mutual Coherence based Technique

- 1: **Input:** Matrix C of size $M \times N$, Matrix V of size $N \times N$, $\vec{\mu}$ vector of zeros of length M
- 2: **for** $i = 1$ to M **do**
- 3: Step 1: New Row in Output Matrix := Perform Linear Combination of Rows of V weighted by weights from i^{th} row of C .
- 4: Step 2: From this New Row, pick up the Maximum of the Absolute values and assign it to μ_i
- 5: **end for**
- 6: Sort the $\vec{\mu}$ vector in Ascending order, preserving the index order.
- 7: Let m be the desired subset size, e.g., $m = 500$.
- 8: Create an array idx_mix containing the first m elements from the sorted index vector.
- 9: Initialize a vector y with m elements, and a matrix C with m rows and N columns, all initialized to zeros.
- 10: **for** $i = 1$ to m **do**
- 11: $y(i) \leftarrow X(idx_mix(i))$
- 12: $C(i, idx_mix(i)) \leftarrow 1$
- 13: **end for**

The rationale of this choice is that **nodes** which have **Least Largest Energy Difference** with some other node, are picked up.

In Simpler terms, that Entity will be preferred to be picked up whose Neighbourhood, is most likely of similar signal values Or in some sense Less Potential Difference from Neighbourhood.



Among Node $\in \{2, 3\}$, $\mu_2 \geq \mu_3$

So Node 3 will be preferred over Node 2

*Good Potential for even Greedier Version than this

A QUICK NOTE Hyper-parameter Selection is Done Based on Cross Validation

Kemeny + MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	6210.7335	0.3515	1	6406.0586	0.2551	195.33	0.10
300	10	6227.8715	0.3317	1	6408.6848	0.2306	180.81	0.10
350	10	6213.6477	0.3223	1	6422.9220	0.2106	209.27	0.11
370	10	6274.5631	0.3183	1	6333.8411	0.2026	59.28	0.12
400	1	6471.0417	0.1775	0.1	6603.6012	0.1745	132.56	0.00
450	1	6375.1552	0.1583	1E-09	6592.0061	0.1510	216.85	0.01
500	1E-08	6332.5379	0.1000	1E-09	6338.8620	0.0999	6.32	0.00
AVERAGE			0.2514			0.1892	142.92	0.06

Lambda Chosen which
Minimises Cross Validation Error

Lambda Chosen which
Minimises RMSE

Difference due to
CHOICE of Lambda

Node Degree + MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	1E-07	7436.8451	0.2599	1	7527.4206	0.2553	90.58	0.00
300	1E-09	7394.9857	0.2348	1	7504.9050	0.2297	109.92	0.01
350	10	7557.9258	0.3227	1	7634.3165	0.2099	76.39	0.11
370	1E-09	7330.3096	0.2047	1	7408.8402	0.2033	78.53	0.00
400	1E-09	7275.3508	0.1737	0.01	7327.7095	0.1735	52.36	0.00
450	1	7423.3918	0.1588	1E-09	7461.1783	0.1521	37.79	0.01
500	1E-09	7286.5961	0.0998	1E-09	7286.5961	0.0998	0.00	0.00
AVERAGE			0.2078			0.1891	63.65	0.02

Lambda Chosen which
Minimises Cross Validation Error

Lambda Chosen which
Minimises RMSE

Difference due to
CHOICE of Lambda

MC + Tuning Lambda with Adjusted Cross Validation Subset								
Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	3007.4391	0.3494	1	3037.8568	0.2559	30.42	0.09
300	10	3022.3485	0.3316	1	3068.4093	0.2283	46.06	0.10
350	10	3041.0553	0.3219	1	3145.0702	0.2098	104.01	0.11
370	10	3113.4056	0.3184	0.1	3374.6716	0.1968	261.27	0.12
400	10	3438.7900	0.3153	0.1	3655.5488	0.1726	216.76	0.14
450	10	3436.4082	0.3104	1E-09	3632.9500	0.1510	196.54	0.16
500	10	3500.3987	0.3031	1E-09	3706.7796	0.0999	206.38	0.20
AVERAGE			0.3214			0.1877	151.63	0.13

Lambda Chosen which
Minimises Cross Validation Error

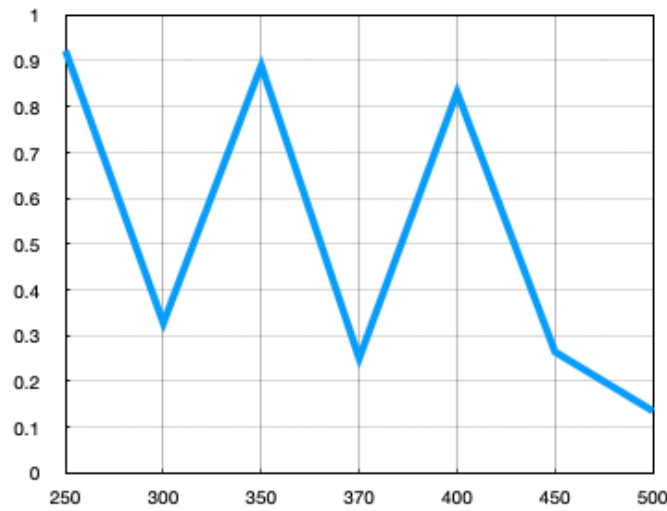
Lambda Chosen which
Minimises RMSE

Difference due to
CHOICE of Lambda

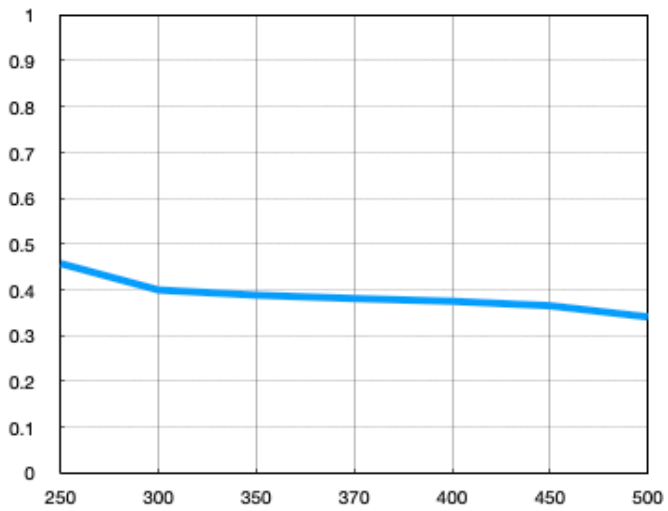
BMC-Measles: Node Granularity H+V

Measurements vs RMSE

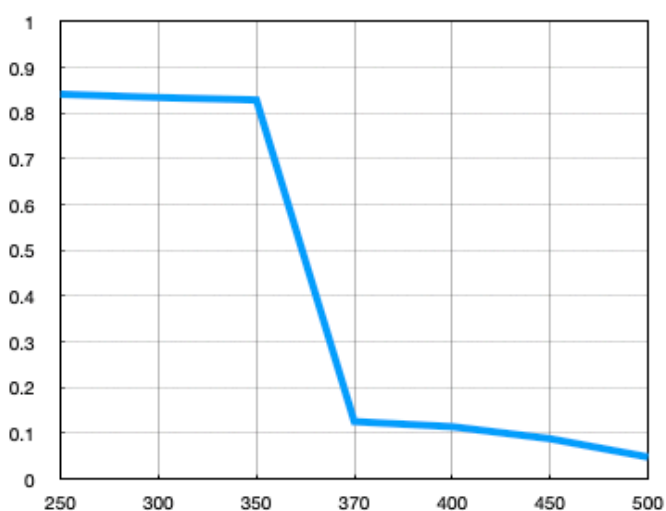
Measurements	Random	Node Degree	Kemeny	MC	AC	Node Degree + MC	Kemeny + MC	Node Degree + AC	Kemeny + AC	True Greedy MC
250	0.9222	0.4568	0.8407	0.2559	0.9037	0.2605	0.2595	0.9048	0.9078	0.9154
300	0.3267	0.3991	0.8334	0.2283	0.9071	0.2297	0.2371	0.9068	0.9046	0.2840
350	0.8891	0.3878	0.8281	0.2137	0.9054	0.2135	0.2159	0.9050	0.9011	0.2791
370	0.2498	0.3806	0.1255	0.1975	0.3865	0.2047	0.2066	0.9034	0.9007	0.2691
400	0.8310	0.3744	0.1145	0.1728	0.3625	0.1739	0.1748	0.3625	0.3618	0.2231
450	0.2639	0.3648	0.0881	0.1513	0.3483	0.1524	0.1512	0.3482	0.3497	0.2221
500	0.1348	0.3403	0.0481	0.1000	0.3222	0.0999	0.0999	0.3222	0.3222	0.1751



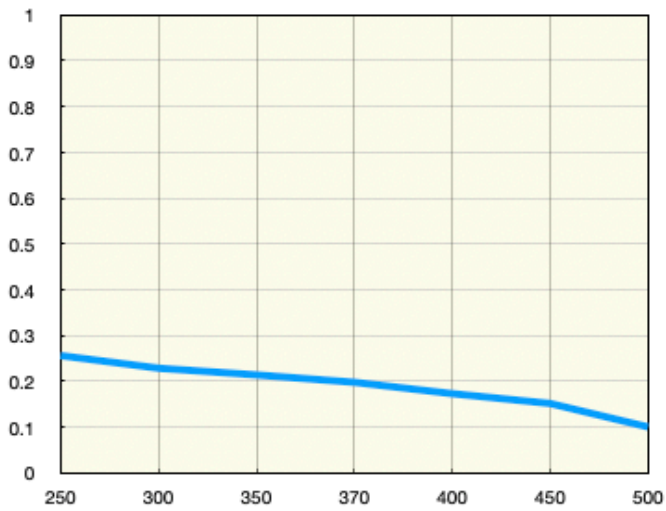
Random



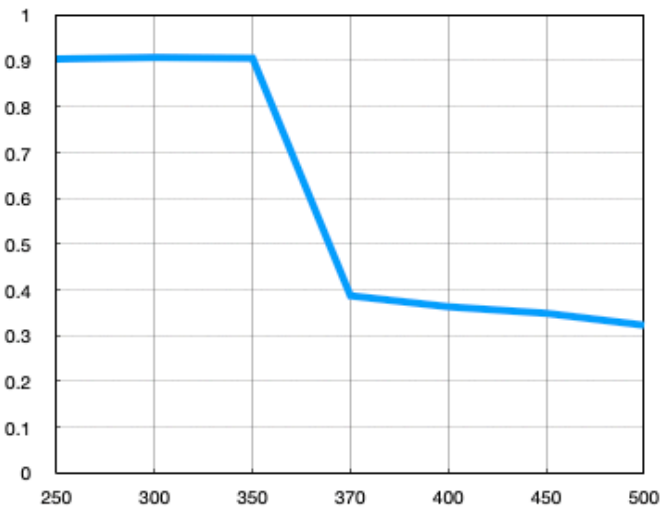
Node Degree



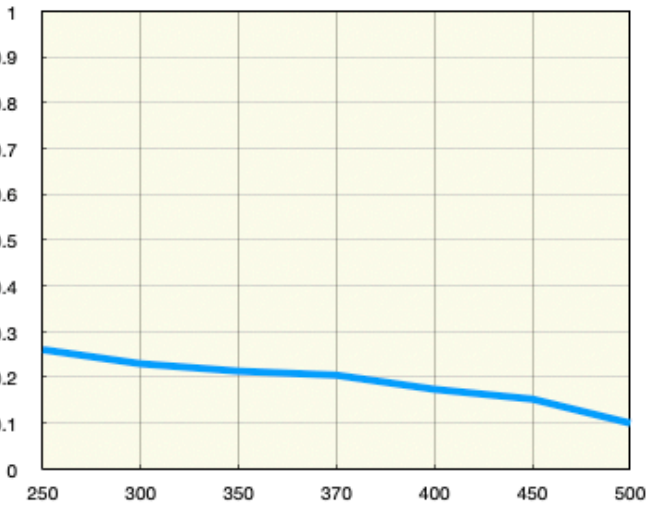
Kemeny



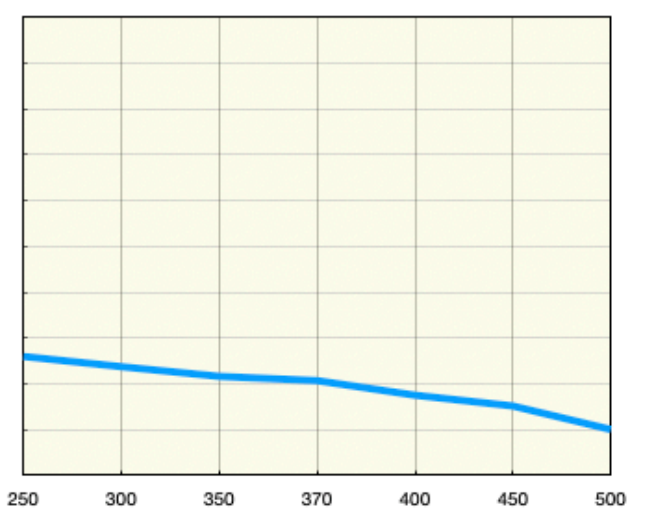
MC



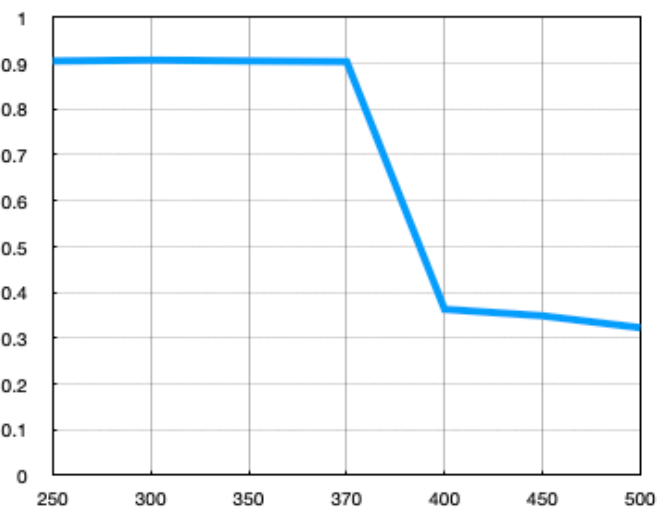
AC



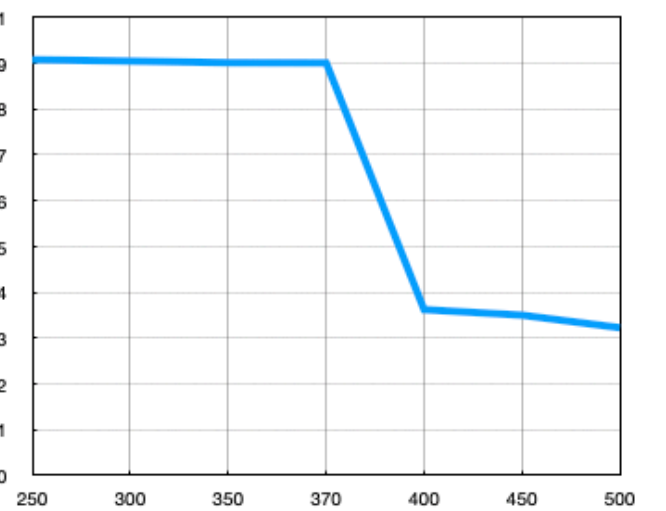
Node Degree + MC



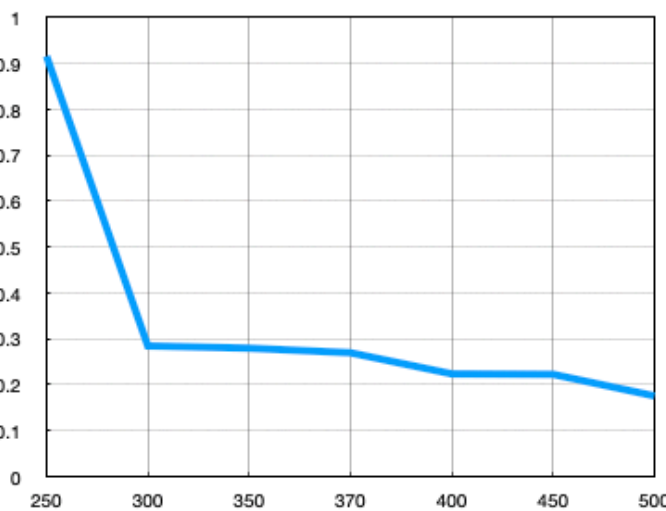
Kemeny + MC



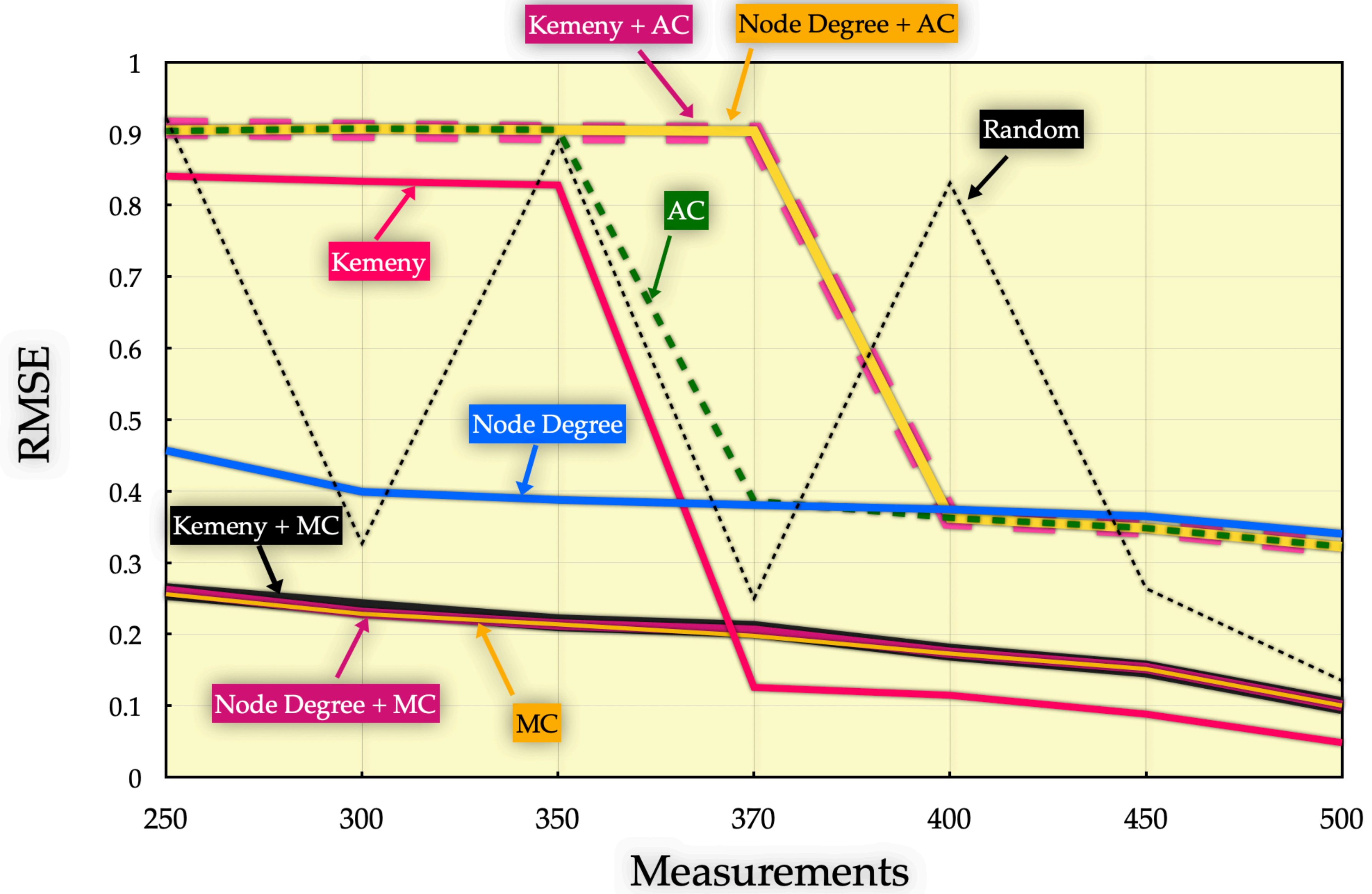
Node Degree + AC

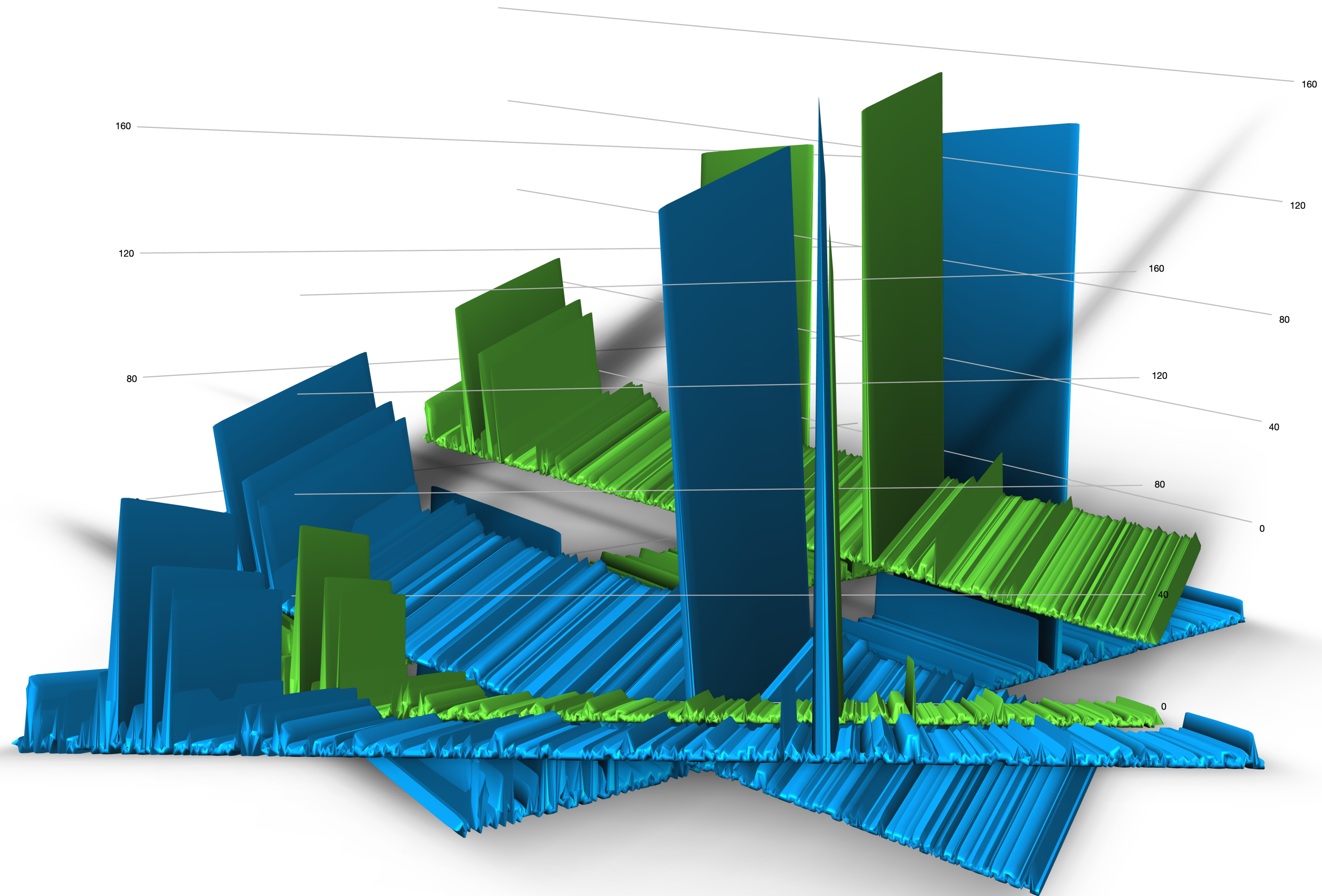


Kemeny + AC



True Greedy MC





Random

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	56.0792	7.269E-11	154.7108	1.1086
300	7.0395	5.794E-11	38.9384	0.8353
350	52.1276	2.354E-11	154.2275	0.8365
370	4.1162	3.357E-09	18.5366	0.8109
400	45.5331	5.119E-11	154.4227	0.5562
450	4.5915	1.150E-10	38.9836	0.3266
500	1.1983	1.481E-10	20.7327	0.1271

MC

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	4.3173	5.748E-10	16.0000	1.0851
300	3.4380	4.160E-10	16.0000	0.8466
350	3.0123	1.088E-10	15.9890	0.5530
370	2.5733	4.470E-11	15.9340	0.4778
400	1.9685	1.958E-11	15.8753	0.3929
450	1.5092	7.733E-09	16.0000	0.2423
500	0.6598	8.802E-09	13.0000	0.0993

Node Degree

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	13.7607	1.536E-10	43.9872	1.4186
300	10.5033	2.725E-10	43.9805	0.9841
350	9.9197	2.400E-09	44.0000	0.8449
370	9.5501	8.988E-09	44.0000	0.7339
400	9.2461	3.810E-11	43.9908	0.6113
450	8.7739	1.561E-09	44.0000	0.7575
500	7.6384	2.460E-12	43.4076	0.3191

Node Degree + MC

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	4.4734	7.420E-11	15.9927	0.9654
300	3.4785	6.051E-10	16.0000	0.8646
350	3.0063	1.314E-10	15.9127	0.5614
370	2.7629	4.019E-11	15.9416	0.5043
400	1.9935	2.570E-09	15.9958	0.3979
450	1.5325	1.582E-08	16.0000	0.2537
500	0.6575	1.176E-09	13.0000	0.0969

Kemeny

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	46.6021	6.780E-11	154.2739	0.9199
300	45.7964	4.694E-12	154.0280	0.7425
350	45.2192	5.520E-11	154.1698	0.6026
370	1.0389	2.961E-09	8.0611	0.3615
400	0.8639	1.292E-10	8.0480	0.2873
450	0.5117	2.170E-08	5.4665	0.1853
500	0.1529	7.730E-12	5.4924	0.0605

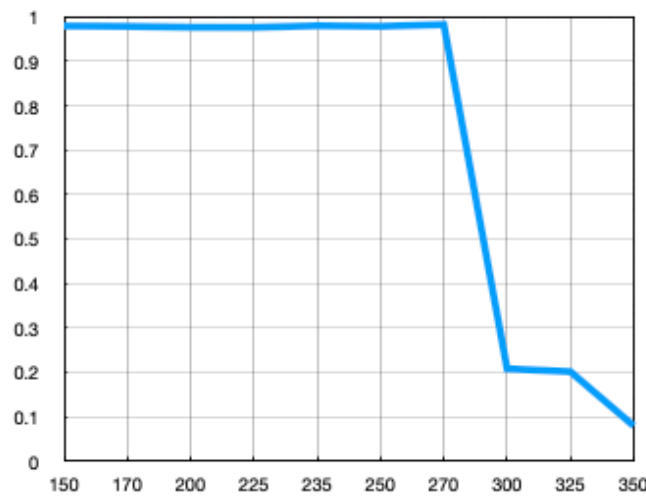
Kemeny + MC

Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	4.4407	7.121E-11	15.8394	0.9356
300	3.7075	2.030E-12	15.9926	0.6970
350	3.0742	9.500E-12	15.8912	0.5666
370	2.8160	2.410E-08	16.0000	0.5077
400	2.0145	1.185E-10	15.8651	0.4095
450	1.5081	1.343E-08	16.0000	0.2422
500	0.6581	1.404E-10	12.9934	0.0993

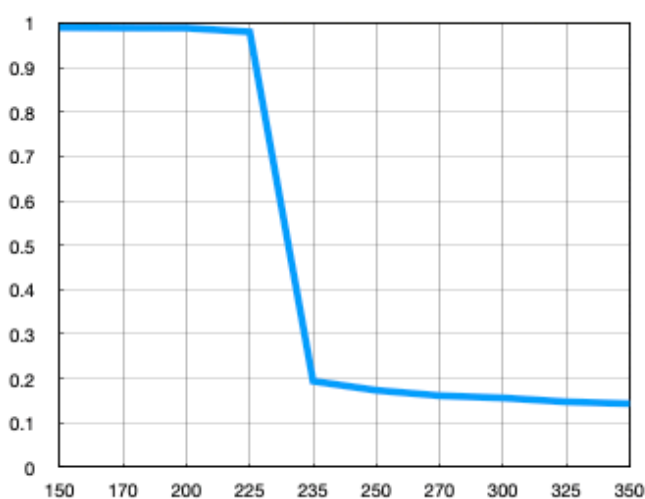
BMC-Measles: Node Granularity H

Measurements vs RMSE 407 Nodes

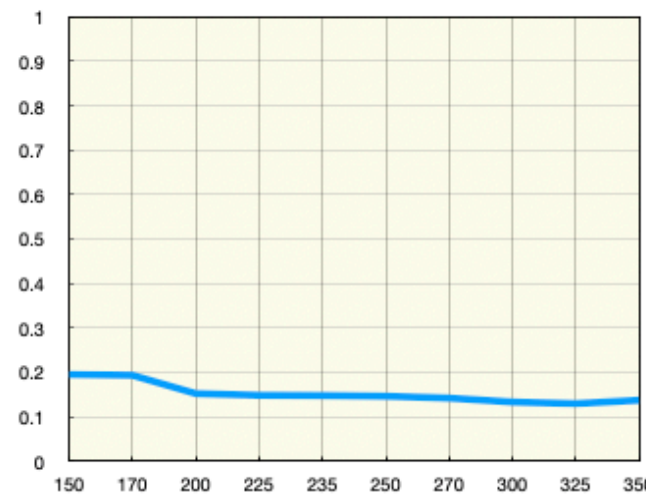
Measurements	~Measurements in %	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
150	37	0.9797	0.9900	0.1953	0.1983	0.1989
170	42	0.9789	0.9892	0.1933	0.1965	0.1951
200	49	0.9769	0.9889	0.1520	0.1512	0.1519
225	55	0.9767	0.9803	0.1482	0.1512	0.1500
235	58	0.9802	0.1939	0.1476	0.1478	0.1466
250	61	0.9790	0.1732	0.1461	0.1463	0.1454
270	66	0.9830	0.1611	0.1419	0.1425	0.1416
300	74	0.2079	0.1560	0.1330	0.1359	0.1330
325	80	0.2016	0.1473	0.1294	0.1298	0.1293
350	86	0.0795	0.1431	0.1371	0.1261	0.1371



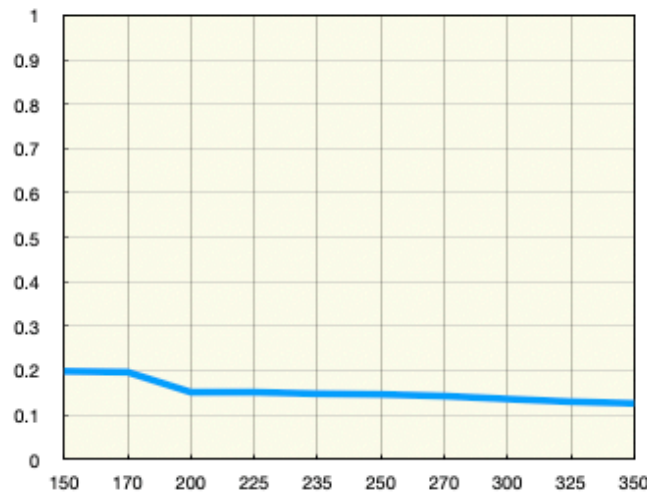
Node Degree



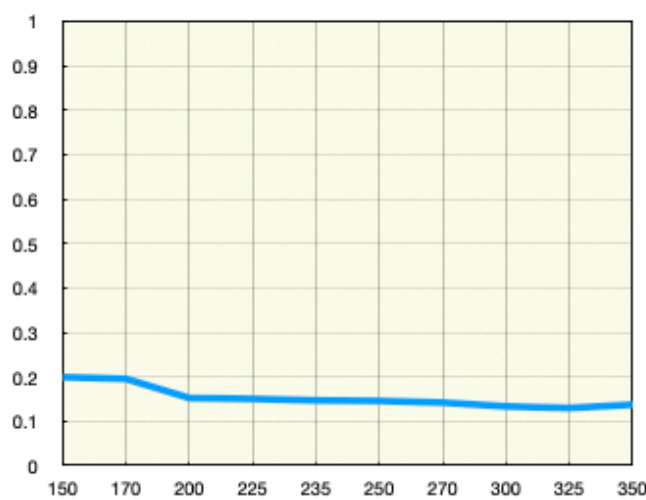
Kemeny



MC



Node Degree + MC



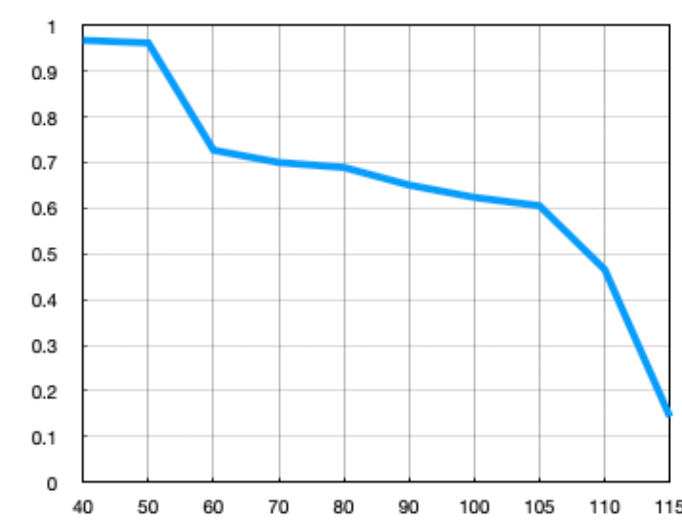
Kemeny + MC

BMC-Measles: Node Granularity V

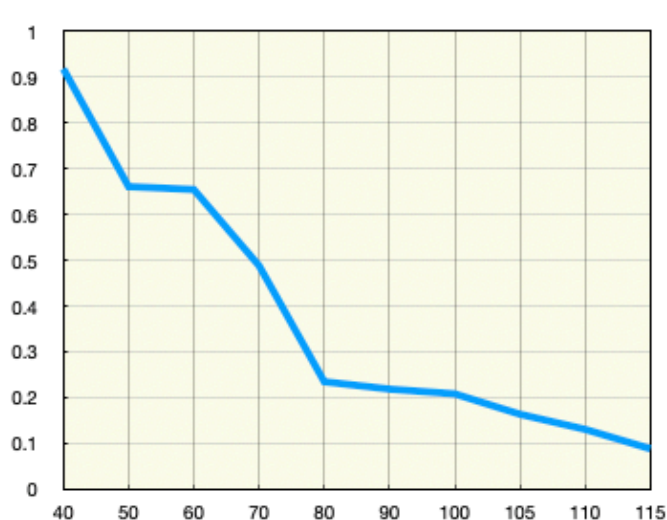
Measurements vs RMSE

126 Nodes

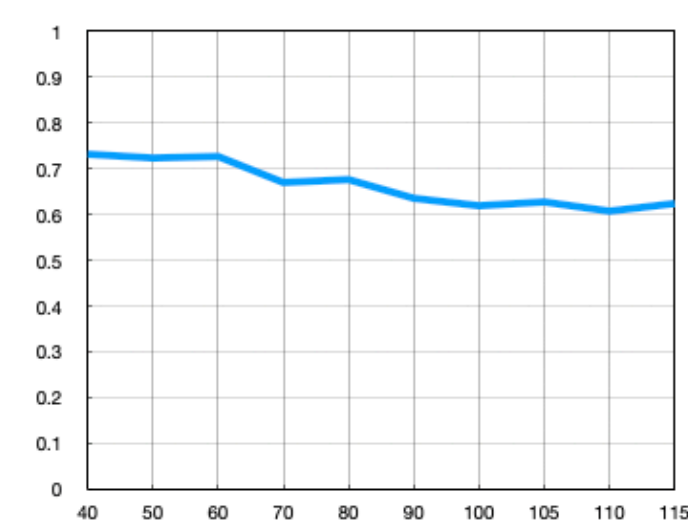
Measurements	Measurements %	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
40	32	0.9666	0.9181	0.7315	0.8752	0.8316	0.4256
50	40	0.9609	0.6605	0.7231	0.7246	0.7157	0.4261
60	48	0.7264	0.6542	0.7265	0.7174	0.7101	0.4249
70	56	0.6995	0.4877	0.6696	0.6860	0.6960	0.4260
80	63	0.6886	0.2341	0.6759	0.6688	0.6742	0.3341
90	71	0.6501	0.2179	0.6351	0.6570	0.6543	0.2822
100	79	0.6232	0.2076	0.6188	0.6332	0.5635	0.2294
105	83	0.6048	0.1630	0.6270	0.6259	0.5883	0.2324
110	87	0.4655	0.1299	0.6069	0.6069	0.6263	0.2428
115	91	0.1437	0.0875	0.6235	0.6233	0.6256	0.2294



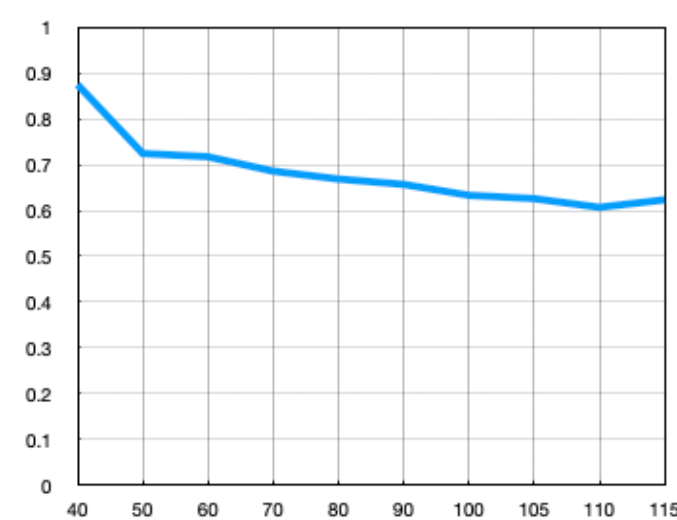
Node Degree



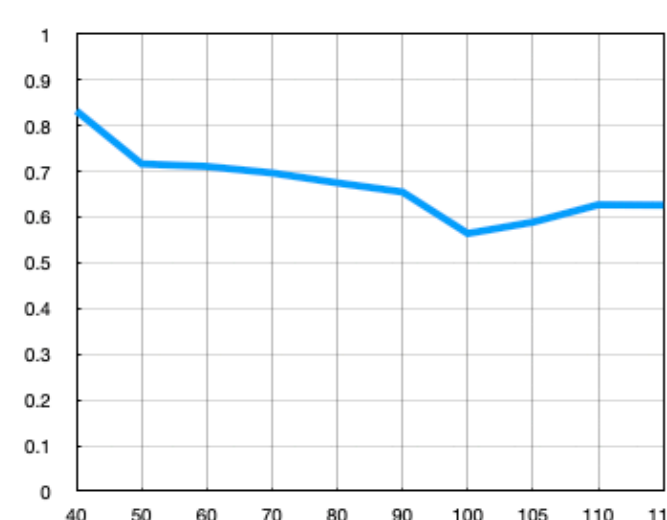
Kemeny



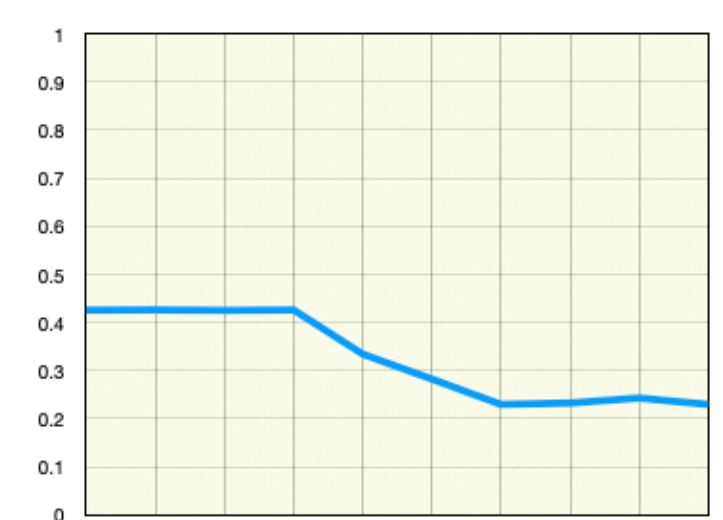
MC



Node Degree + MC



Kemeny + MC

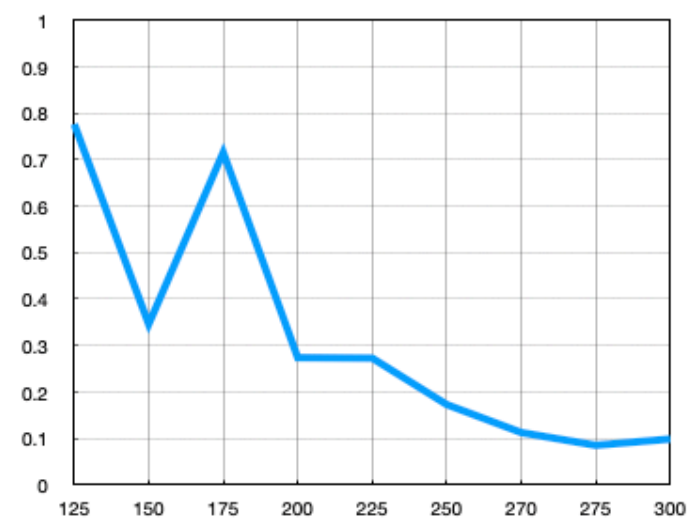


True Greedy MC

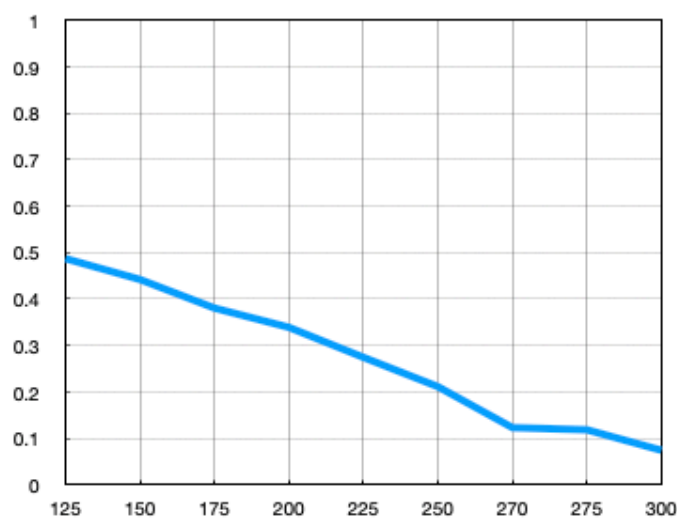
WHO-Measles: Node Granularity H+V

Measurements vs RMSE

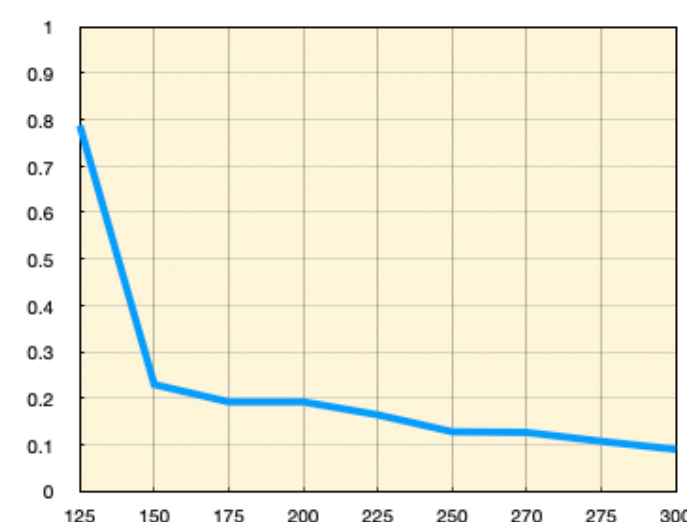
Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
125	0.7766	0.4869	0.7874	0.3835	0.3870	0.3791	0.6896
150	0.3458	0.4420	0.2296	0.3655	0.3719	0.3207	0.6732
175	0.7155	0.3803	0.1921	0.3616	0.3617	0.3463	0.6416
200	0.2736	0.3389	0.1921	0.3593	0.3599	0.3463	0.6013
225	0.2726	0.2742	0.1643	0.2687	0.2689	0.2687	0.5692
250	0.1726	0.2111	0.1278	0.1157	0.1175	0.1060	0.5355
270	0.1123	0.1229	0.1264	0.0918	0.0918	0.0918	0.5346
275	0.0849	0.1183	0.1072	0.0876	0.0916	0.0876	0.5337
300	0.0984	0.0741	0.0899	0.0655	0.0655	0.0655	0.5283



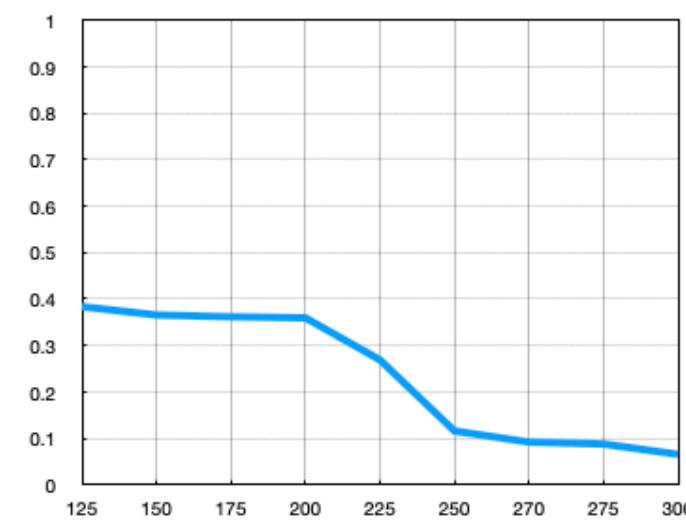
Random



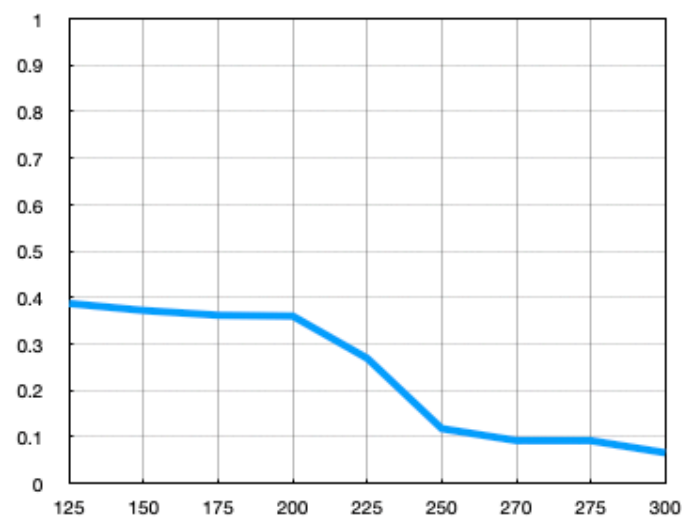
Node Degree



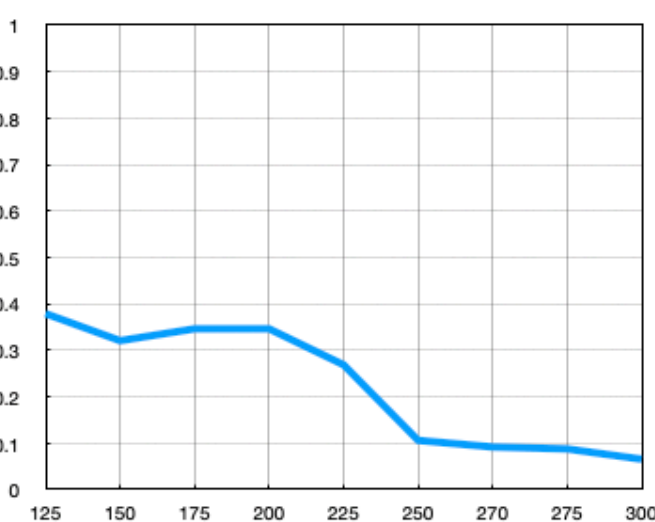
Kemeny



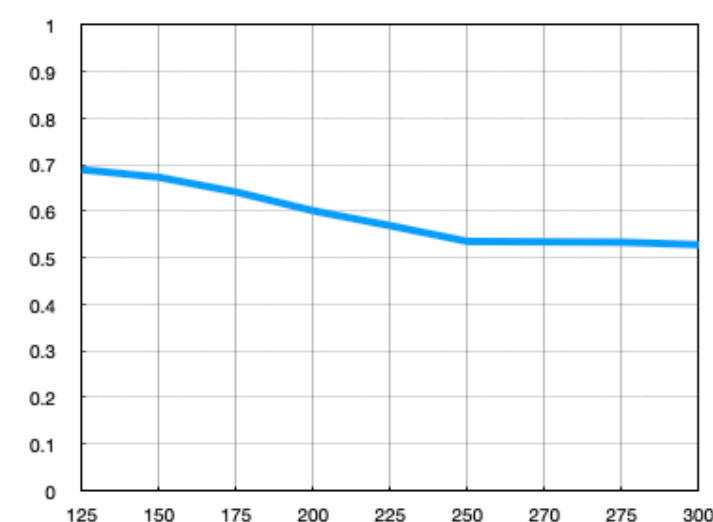
MC



Node Degree + MC



Kemeny + MC



True Greedy MC

WHO-Measles: Node Granularity H+V

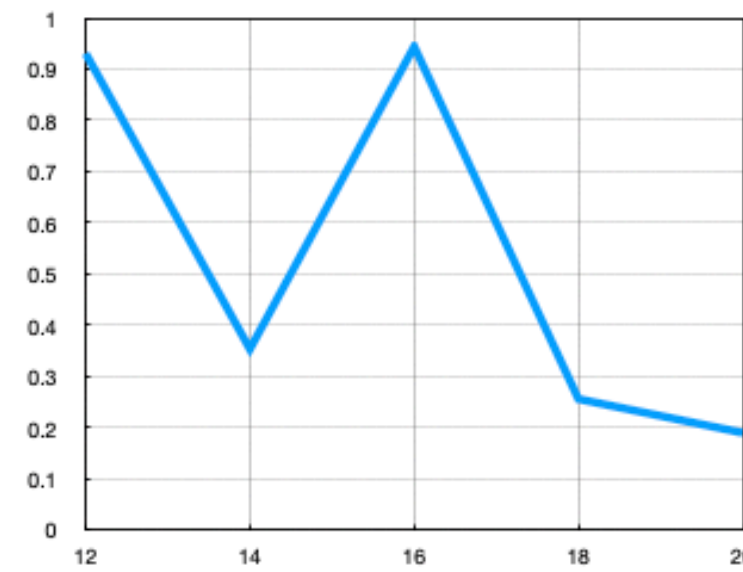
Measurements vs MSE

Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
125	3330.92	1309.60	3424.24	812.14	827.18	793.61	2626.57
150	660.26	1078.76	2994.78	737.87	763.99	567.97	2503.15
175	2827.24	798.96	291.04	722.03	722.38	662.24	2273.32
200	413.51	634.15	203.89	713.09	715.24	662.28	1996.65
225	410.38	415.36	149.08	398.79	399.34	398.81	1789.32
250	164.49	246.23	90.23	73.93	76.26	62.08	1583.54
270	69.70	83.43	88.20	46.52	46.53	46.52	1578.44
275	39.76	77.28	63.51	42.40	46.38	42.40	1573.34
300	53.52	30.33	44.61	23.67	23.67	23.67	1541.75

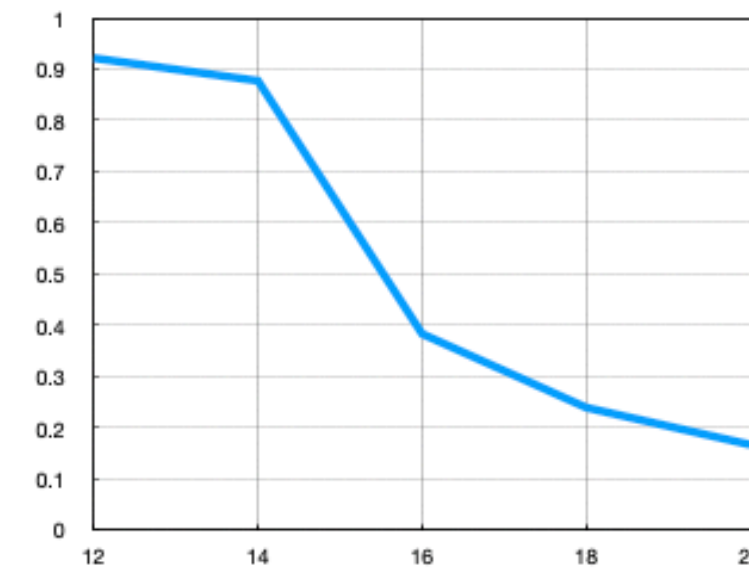
WHO-Measles: Ward Granularity

Measurements vs RMSE

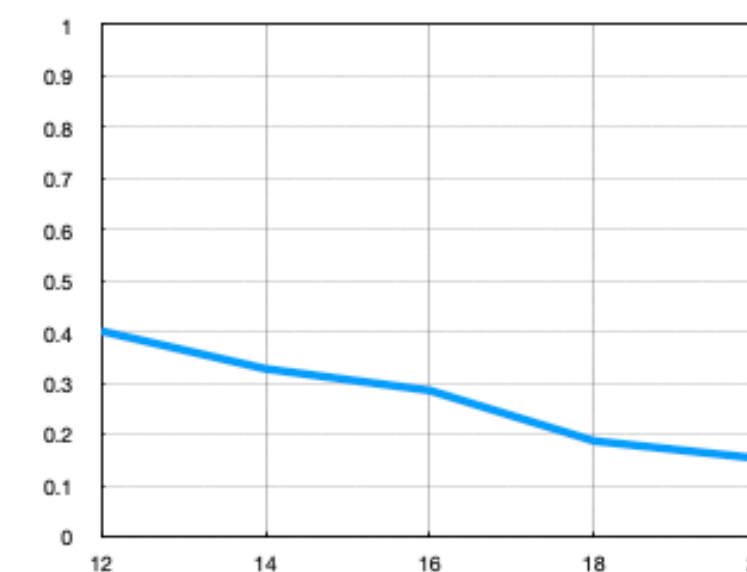
Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
12	0.9305	0.9214	0.4023	0.2233	0.3135	0.2233	0.9453
14	0.3532	0.8771	0.3284	0.2103	0.2296	0.2103	0.9415
16	0.9436	0.3830	0.2862	0.1989	0.2080	0.1989	0.9626
18	0.2557	0.2389	0.1876	0.1054	0.2027	0.1054	0.1030
20	0.1897	0.1664	0.1536	0.1084	0.0596	0.1084	0.1321



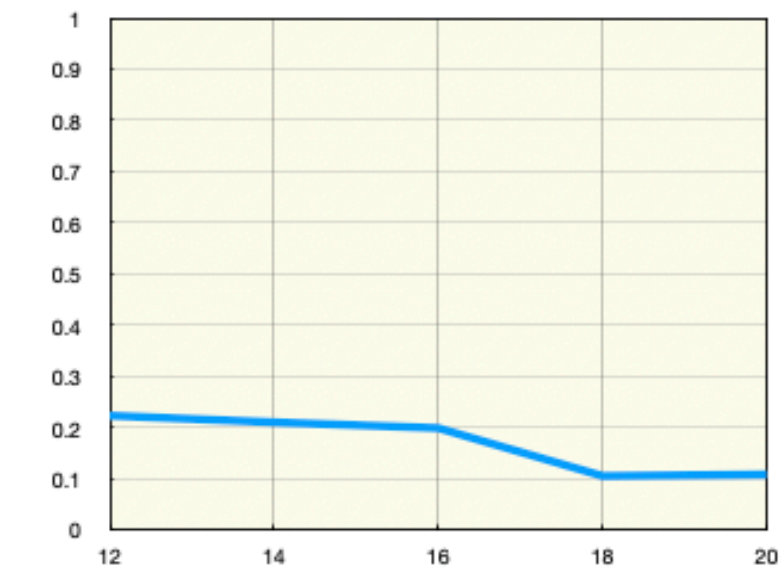
Random



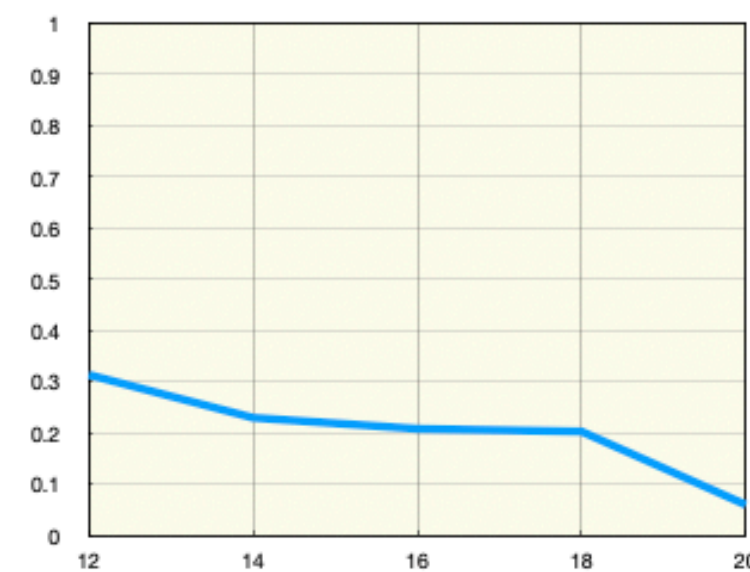
Node Degree



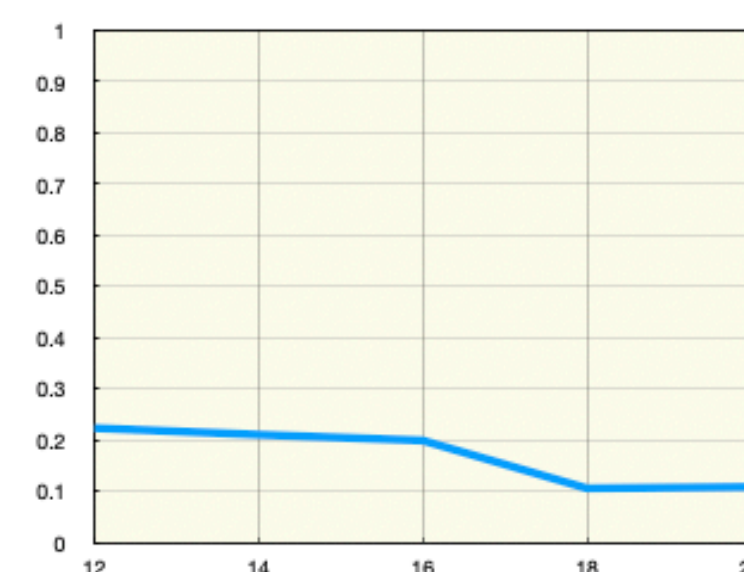
Kemeny



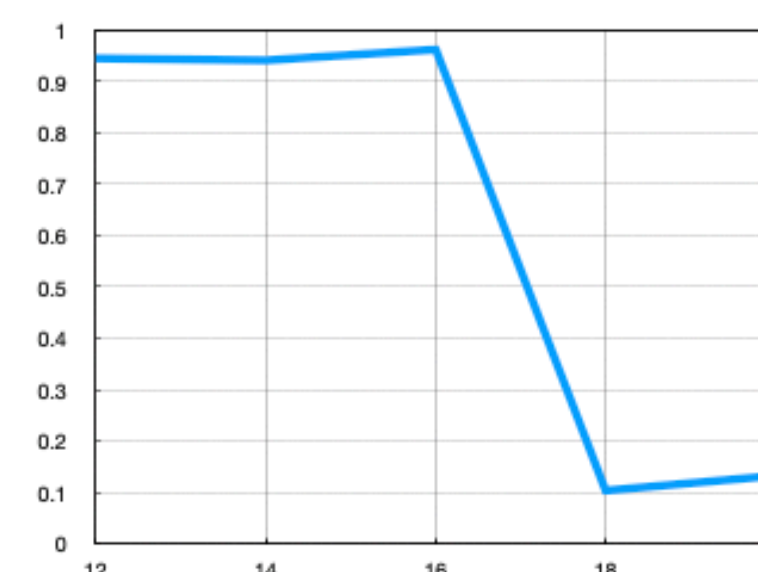
MC



Node Degree + MC



Kemeny + MC



True Greedy MC

US-Covid-19: County + Province Granularity

Model Setup

Dataset

Mobility data



US Govt Official Census

Used to build
Contact Tracing
Map

No Lat/Long Data. Here
Used Avg Commuters from
Place A to B Symmetrically

+

Covid19 Data From JHU

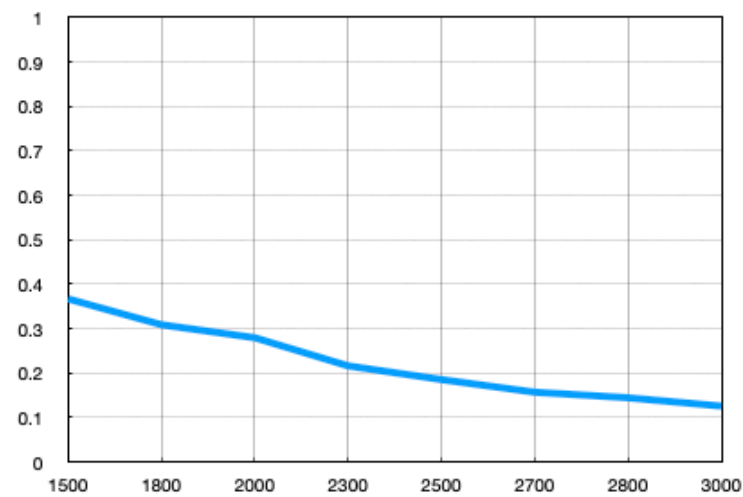


Used for assigning
Signals values to
Entities as # Cases
Registered

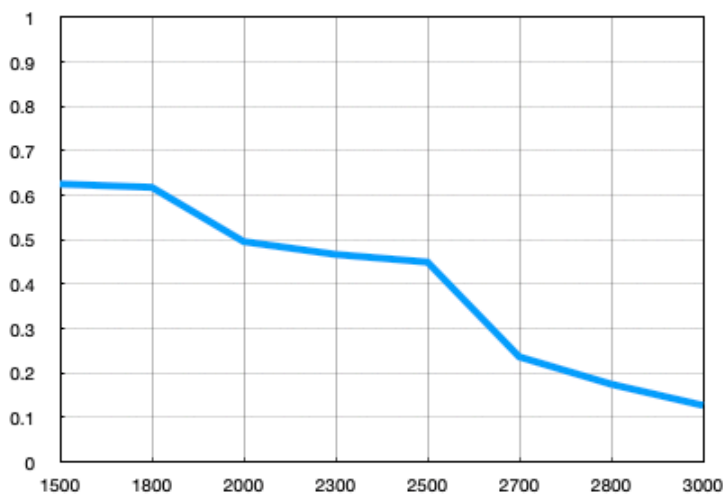
US-Covid-19: County Granularity

Measurements vs RMSE					
Measurements	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
1500	0.3662	0.6245	0.9938	0.7078	0.9943
1800	0.3080	0.6171	1.0007	0.6825	1.0007
2000	0.2788	0.4950	0.9979	0.6616	0.9976
2300	0.2156	0.4662	0.9472	0.5996	0.9469
2500	0.1848	0.4490	0.9328	0.5710	0.9328
2700	0.1563	0.2358	1.1031	0.8767	1.1031
2800	0.1439	0.1744	1.1361	0.7462	1.1361
3000	0.1251	0.1266	0.5341	0.3686	0.5341

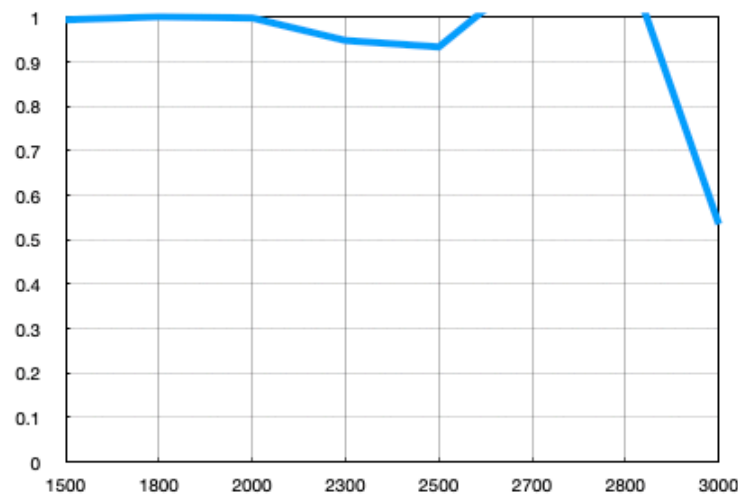
Node Degree



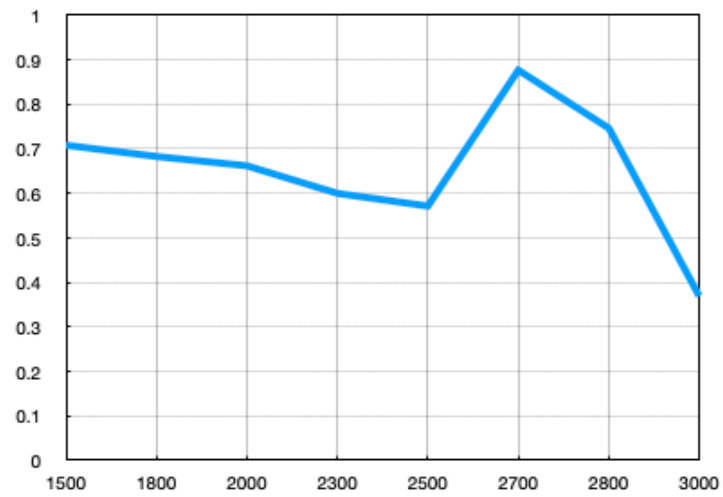
Kemeny



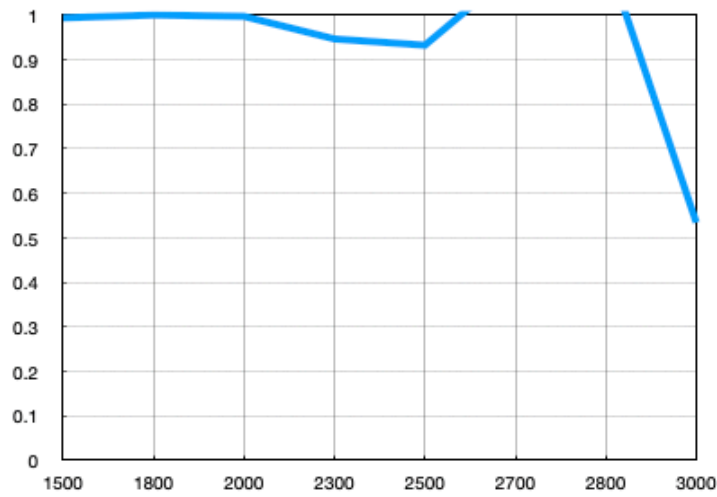
MC



Node Degree + MC



Kemeny + MC



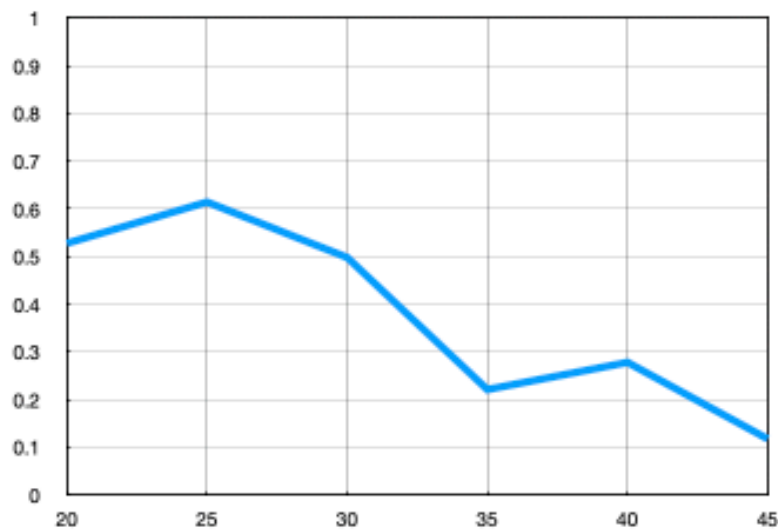
US DATASET
COUNTY GRANULARITY
FISTA BASED

US-Covid-19: Province Granularity

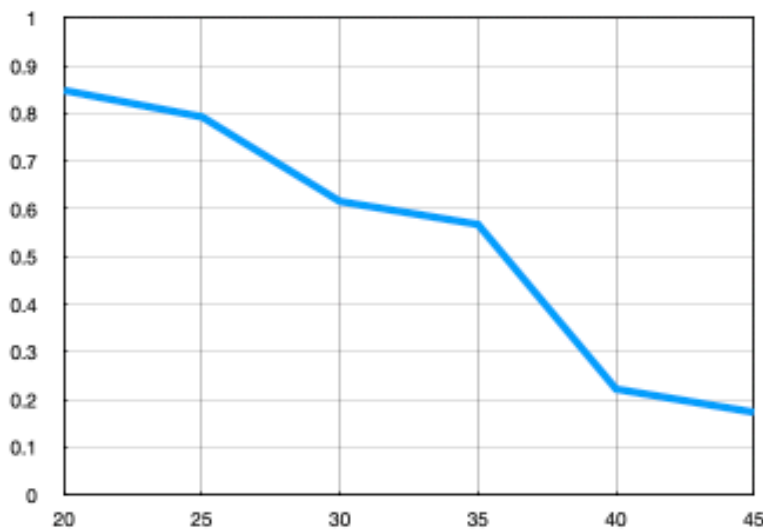
Measurements vs RMSE

Measurements	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
20	0.5278	0.8484	0.7125	0.7958	0.6649
25	0.6142	0.7928	0.7084	0.6769	0.6476
30	0.4980	0.6155	0.3963	0.7018	0.4712
35	0.2211	0.5673	0.3886	0.3873	0.3818
40	0.2781	0.2222	0.2269	0.2171	0.3724
45	0.1176	0.1734	0.1804	0.1727	0.2228

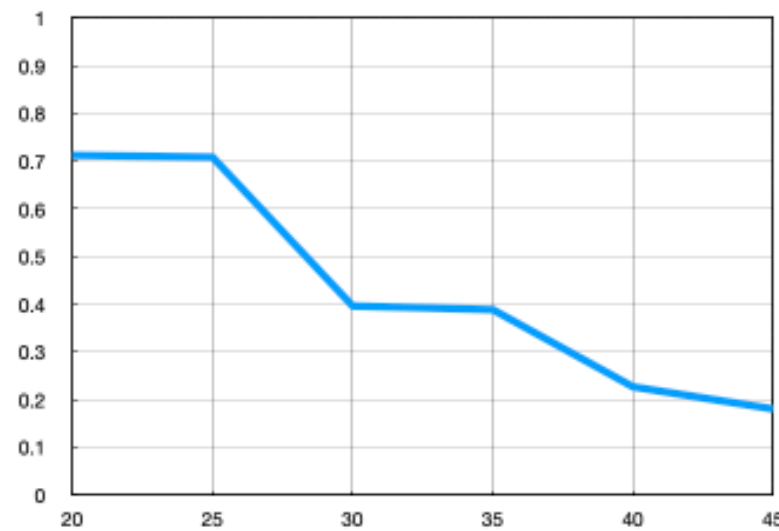
Node Degree



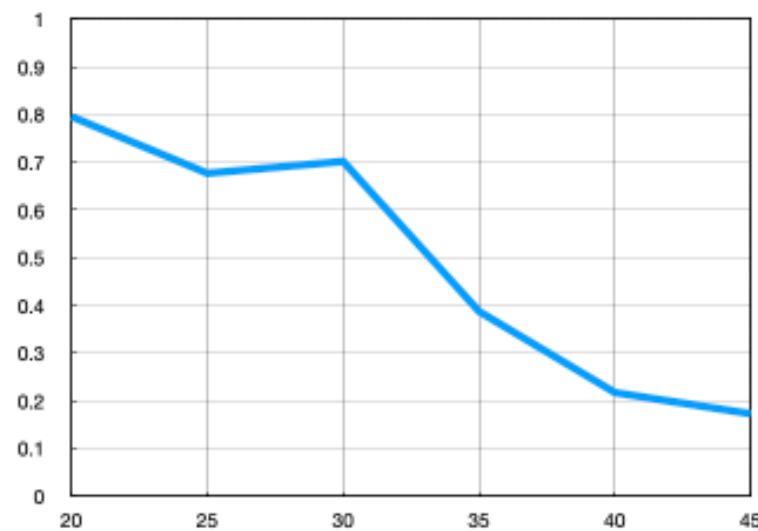
Kemeny



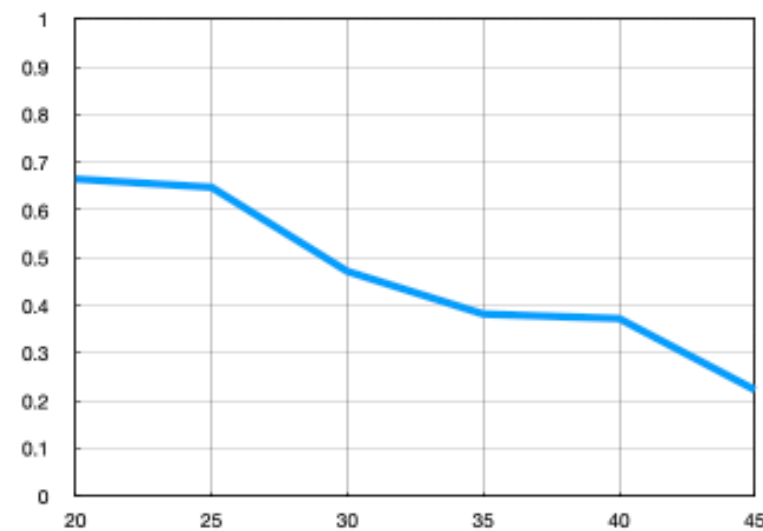
MC



Node Degree + MC



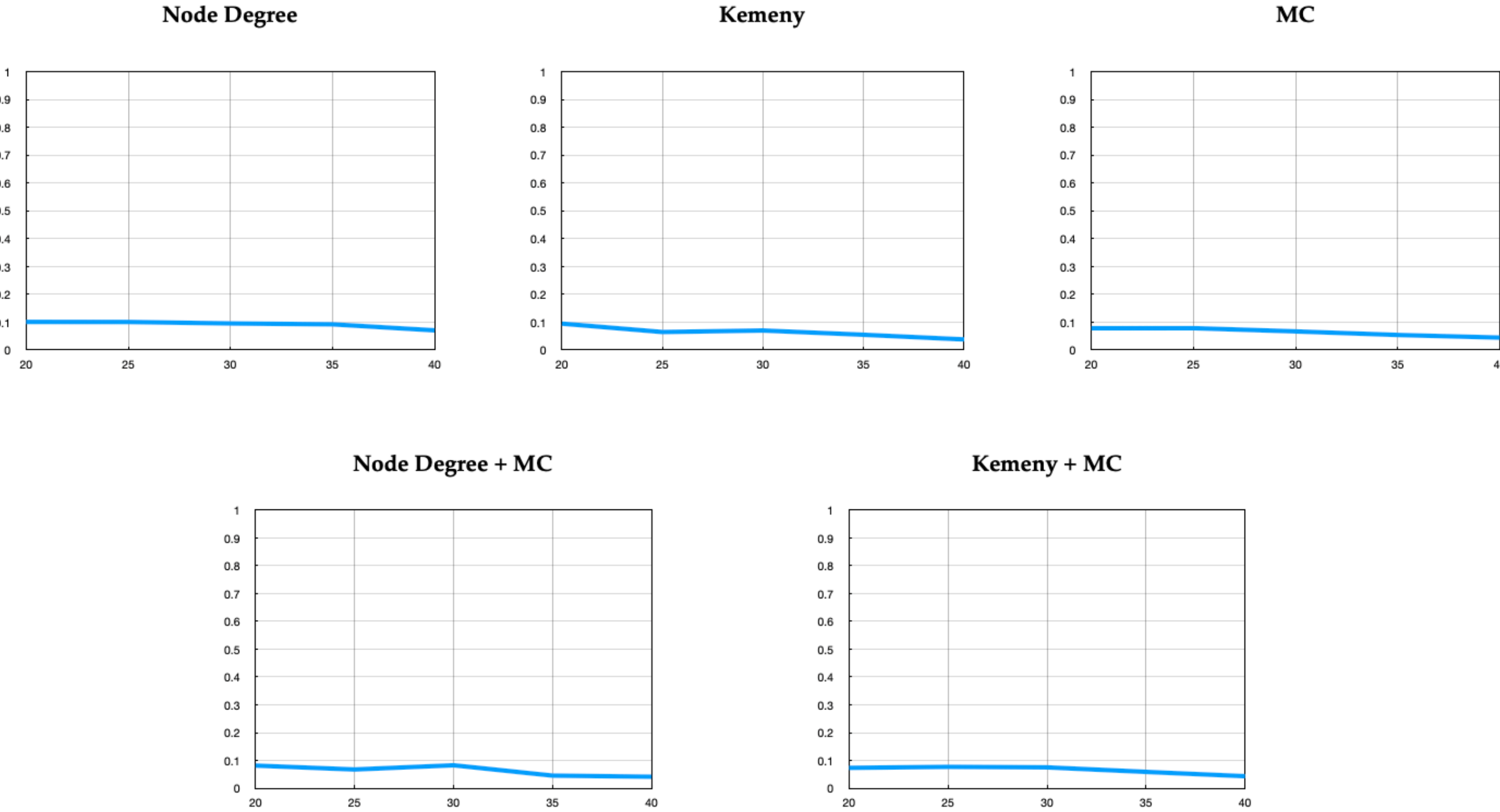
Kemeny + MC



US-Covid-19: Province Granularity (Log Scale Mobility Data)

Measurements vs RMSE

Measurements	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC
20	0.1001	0.0939	0.0773	0.0814	0.0732
25	0.0998	0.0637	0.0776	0.0673	0.0768
30	0.0946	0.0691	0.0659	0.0825	0.0749
35	0.0913	0.0538	0.0531	0.0453	0.0587
40	0.0696	0.0370	0.0435	0.0413	0.0431



Simulation on Synthetically Generated Data Person-To-Person

Subset Selection Technique	Viral Load	Recovery	Model Modification
Random	Yes	No	No
Node Deg+ K+MC	Yes	No	No
Node Deg+ K+MC	No	Yes (Relaxed)	No
Node Deg+ K+MC	No	Yes (Accelerated)	No
Node Deg+ K+MC	No	Yes	Log Weighted Lasso

*Simulated over 500 Nodes wide range of Days upto 500 Days

Graph Learning

Gaussian RBF Kernel, a common Edge Weighing Function Heuristic

When the edge weights are not naturally defined by an application, one common way to define the weight of an edge connecting vertices i and j $W_{i,j}$ is via a thresholded Gaussian kernel weighting function

$$W_{i,j} = \begin{cases} \exp\left(-\frac{[\text{dist}(i, j)]^2}{2\theta^2}\right) & \text{if } \text{dist}(i, j) \leq \kappa \\ 0 & \text{otherwise,} \end{cases}$$

for some parameters θ and κ
 $\text{dist}(i, j)$ may represent a physical distance between vertices i and j , or the Euclidean distance between two feature vectors describing i and j

This is a Heuristic which is commonly applied to represent the edge weights, thus we wished to infer directly from the data.

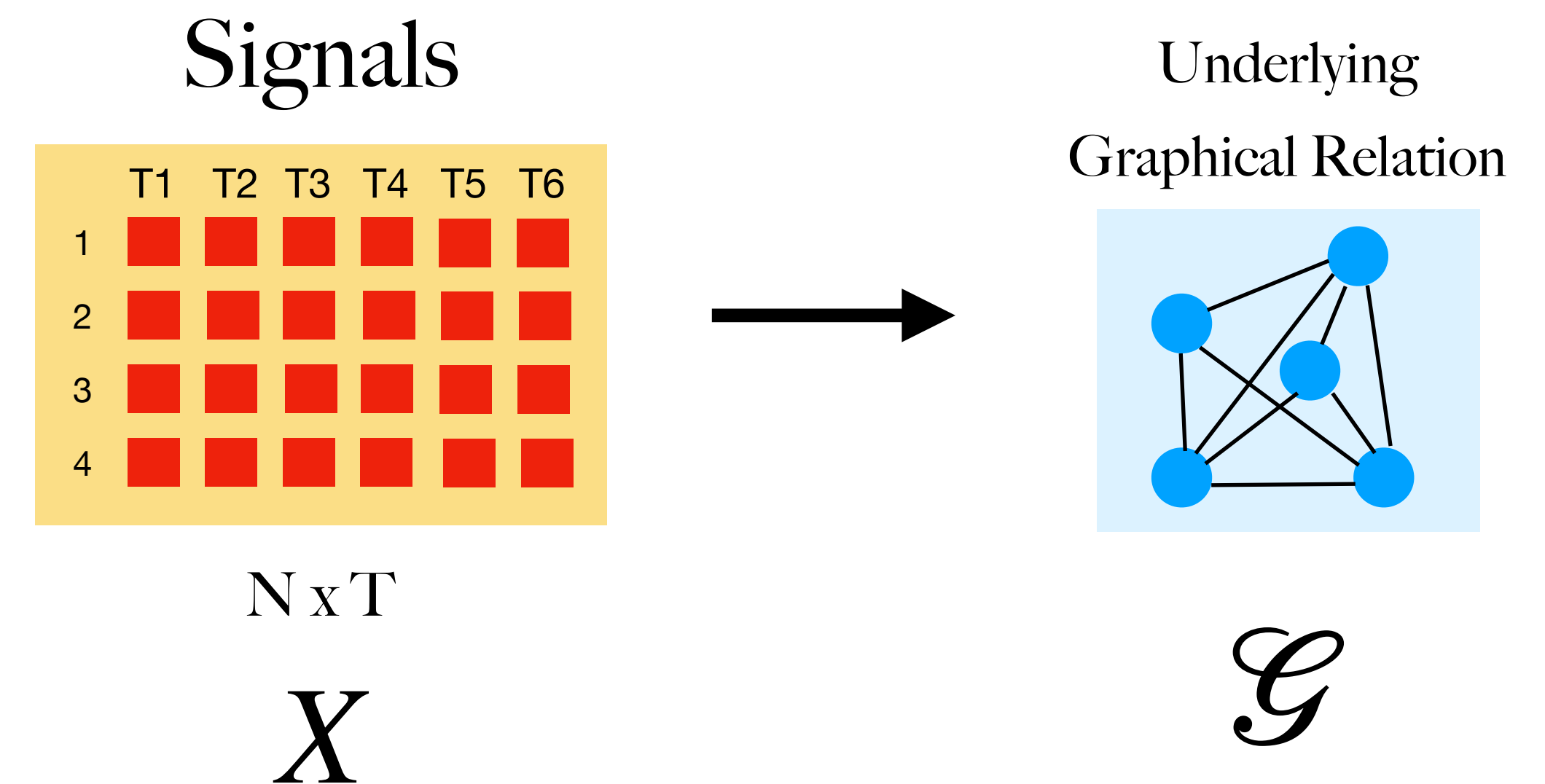
What is Graph Learning Problem ?

When a natural choice of the graph is not readily available from the data sets, it is thus desirable to infer or learn a graph topology from the data.

Problem Formulation

Given T observations for N Entities, we would like to infer the Underlying **Relationship** that Exists within them.

Each Column of the X Matrix Represents a Graph Signal defined over all the Entities.



How in Graphical Models, Edge Weights Were Inferred?

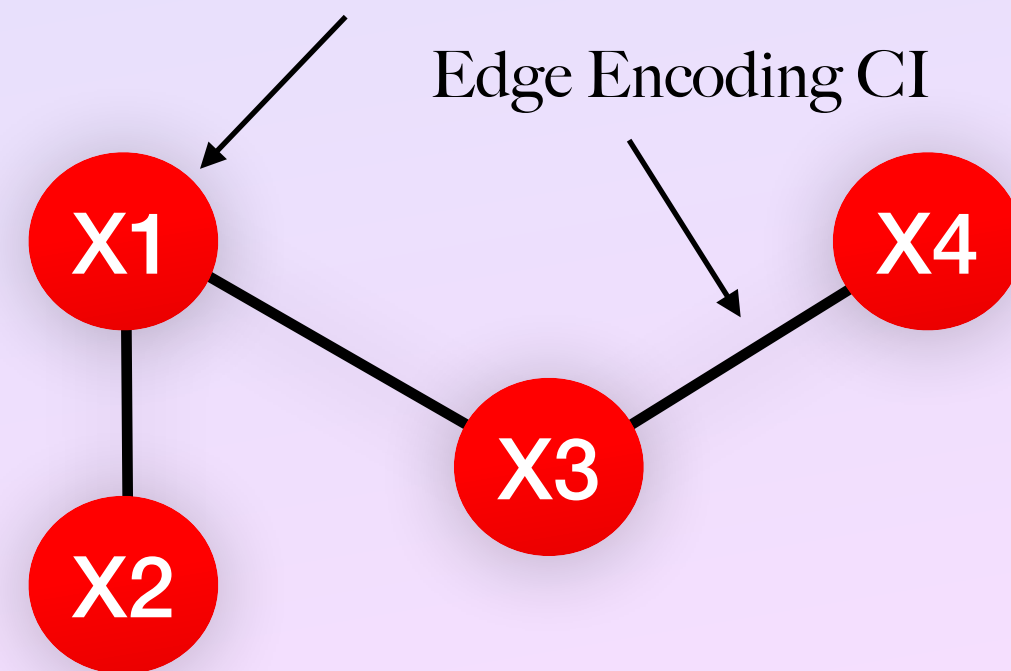
General Philosophy

Toy Graphical Model

Suppose we have 4 R.V.s $X = \{X_1, X_2, X_3, X_4\}$

R.V. as Node/Vertex

Edge Encoding CI



$$P(X) \propto \prod_{i=1}^N \phi_i \prod_{i,j=1, i \neq j}^N \phi_{i,j}$$

Node Potential

Edge Potential

Learning Edges Weights

For Exponential family of Distribution

$$\phi_i = e^{\theta_{i,i} x_{i,i}^2} \quad \phi_{i,j} = e^{\theta_{i,j} x_i x_j}$$

Joint Probability Distribution now becomes

$$P(X) = \frac{1}{Z(\theta)} \cdot e^{\sum \theta_{i,j} x_i x_j + \theta_{i,i} x_i^2}$$

Now Finding the Parameter Matrix θ was a way to infer the relation among Nodes in Graphical Models

$$P(X) = \frac{1}{Z(\theta)} \cdot e^{\sum \theta_{i,j} x_i x_j + \theta_{i,i} x_i^2}$$

Idea

$$P(x) = \frac{|\Theta|^{1/2}}{(2\pi)^{N/2}} \cdot e^{-(1/2) \cdot x' \cdot \Theta \cdot x}$$

$$\log(L(X)) = \sum_{i=1}^N \log(P(x_i))$$

$$\max_{\Theta} \log(L(X))$$

Closed Form Joint Probability Distribution of GMRF ($\mu = 0$)

Gaussian Graphical Model or **GMRF** $\Rightarrow$

$$p(\mathbf{x} | \Theta) = \frac{|\Theta|^{1/2}}{(2\pi)^{N/2}} \exp\left(-\frac{1}{2} \mathbf{x}^T \Theta \mathbf{x}\right)$$

How in Graphical Models, Edge Weights Were Inferred?

Graphical Lasso

Smoothness Term

$$\begin{aligned} &\underset{\Theta, \sigma^2}{\text{maximize}} && \log \det \Theta - \text{tr} \left(\frac{1}{M} \mathbf{X} \mathbf{X}^T \Theta \right) - \rho \|\Theta\|_1, \\ &\text{subject to} && \Theta = \mathbf{L} + \frac{1}{\sigma^2} \mathbf{I}, \mathbf{L} \in \mathcal{L}, \end{aligned}$$

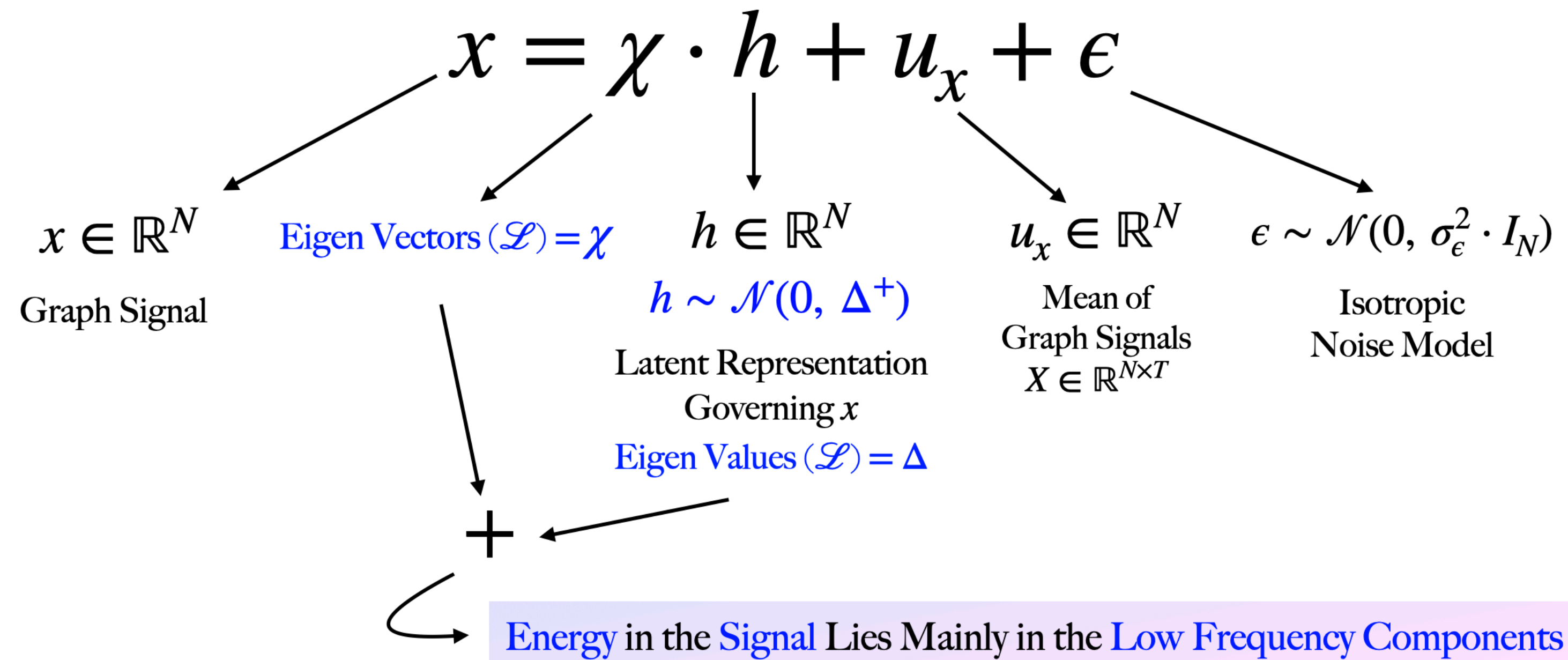
It was desired to learn those kind of Inverse Covariance Matrices, where off Diagonal Elements were ≤ 0

By solving for it, the authors obtain the **Graph Laplacian Matrix**, or equivalently, an **Adjacency Matrix** with **Non-Negative Weights**.

How in GSP, an attempt was Made to Learn the Edge Weights ?

Dong's Approach

They make an **Assumption** that **Graph Signals** x are represented as



Assuming, without loss of Generality

$$u_x = 0$$

We Infer the Following

$$x | h \sim \mathcal{N}(\chi \cdot h, \sigma_\epsilon^2 \cdot I_N)$$

$$\log(P(x | h)) \propto -\|x - \chi \cdot h\|_2^2$$

$$x \sim \mathcal{N}(0, \mathcal{L}^+ + \sigma_\epsilon^2 \cdot I_N)$$

$$h \sim \mathcal{N}(0, \Delta^+)$$

$$\log(P(h)) \propto -h^T \cdot \Delta \cdot h$$

MAP Estimate for h

$$\begin{aligned} h_{map} &= \arg \max_h P(h | x) \\ &= \arg \max_h P(x | h) \cdot P(h) \\ &= \arg \max_h \log(P(x | h)) + \log(P(h)) \\ &= \arg \min_h \|x - \chi \cdot h\|_2^2 + \alpha \cdot h^T \cdot \Delta \cdot h \end{aligned}$$

Change of Variable

$$y = \chi \cdot h \implies h = \chi^T \cdot y$$

$$h_{map} = \arg \min_h \|x - y\|_2^2 + \alpha \cdot y^T \cdot \mathcal{L} \cdot y$$

When the $\mathcal{G}$ is unknown, we can therefore have the following joint optimisation problem in order to infer the $\mathcal{L}$

$$\arg \min_{\mathcal{L}, y} \|x - y\|_2^2 + \alpha \cdot y^T \cdot \mathcal{L} \cdot y$$

How in GSP, an attempt was Made to Learn the Edge Weights ?

Dong's Approach

$$\begin{aligned} \min_{L \in \mathbb{R}^{n \times n}, Y \in \mathbb{R}^{n \times p}} \quad & ||X - Y||_F^2 + \alpha \operatorname{tr}(Y^T L Y) + \beta ||L||_F^2 \\ \text{s.t.} \quad & \operatorname{tr}(L) = n, \\ & L_{ij} = L_{ji} \leq 0, \quad i \neq j, \\ & L \cdot \mathbf{1} = \mathbf{0}, \end{aligned}$$

$$X \in \mathbb{R}^{n \times p}$$

$$Y = \chi \cdot h$$

$y^T \cdot \mathcal{L} \cdot y$: Gives Measure of Smoothness on the Graph

$$\begin{array}{ll} \dim \chi & n \times n \\ \dim h & n \times p \end{array} \quad \because y^T \cdot \mathcal{L} \cdot y = \frac{1}{2} \cdot \sum_{i,j} w_{i,j} \cdot (y(i) - y(j))^2 = 2 \cdot \operatorname{tr}(Y^T \cdot L \cdot Y)$$

Thus here using this approach we learn the Graph Laplacian $\mathcal{L}$

How in GSP, an attempt was Made to Learn the Edge Weights ?

Kalofolias's Approach

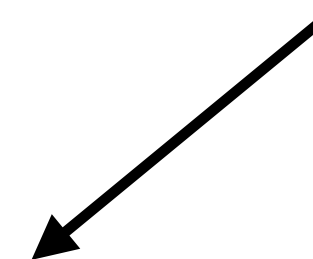
The main Idea is instead of $\mathcal{L}$ we are Learning $\mathcal{W}$

Let us define the *pairwise distances matrix* $Z \in \mathbb{R}_+^{m \times m}$ $Z_{i,j} = \|x_i - x_j\|^2$

Consider the Smoothness term: $\text{tr}(X^\top LX) = \frac{1}{2} \text{tr}(WZ) = \frac{1}{2} \|W \circ Z\|_{1,1}$



$$\underset{W \in \mathcal{W}_m}{\text{minimize}} \quad \|W \circ Z\|_{1,1} - \alpha \mathbf{1}^\top \log(W \mathbf{1}) + \beta \|W\|_F^2$$



Ensures Graph Connectedness



Imposes Graph Sparsity

What is Lacking in All these formulations ?

In all these formulations **Band limited Signals** were being considered i.e. only **certain frequency Components** from the **spectral domain** are responsible for bringing the **smoothness** in the **Graph Signals**.

Ideally this could have been **True**, if all the **signals** were **Generated** from a **Diffusion Process**, but which is certainly not **Pragmatic!**

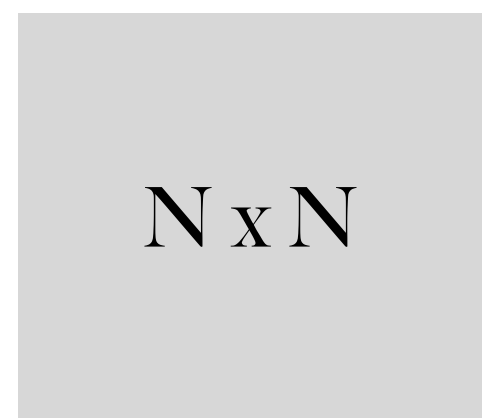
So We Propose Our Formulation

STEP 1

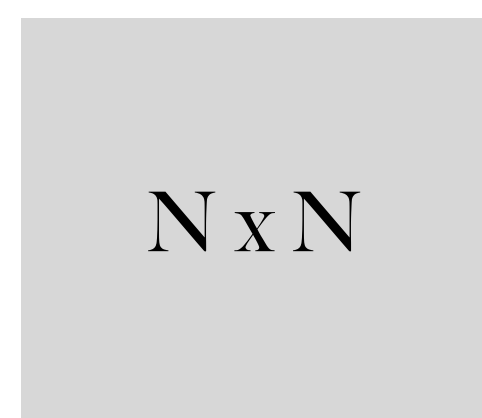
To Obtain Sparse Representation in Spectral Domain

Impose Signal Sparsity

Adjacency Matrix W



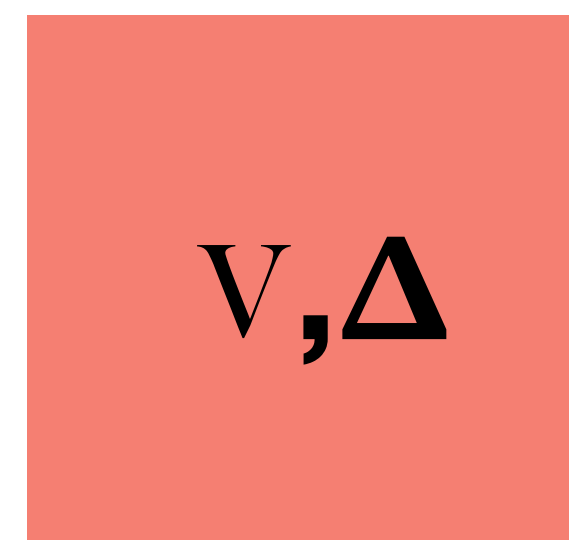
Graph Laplacian L



$$L = D - W$$

It's Eigen Vectors
Captures the Energy in
the Signals

Transforming
Basis Vectors



$N \times N$

Orthonormal

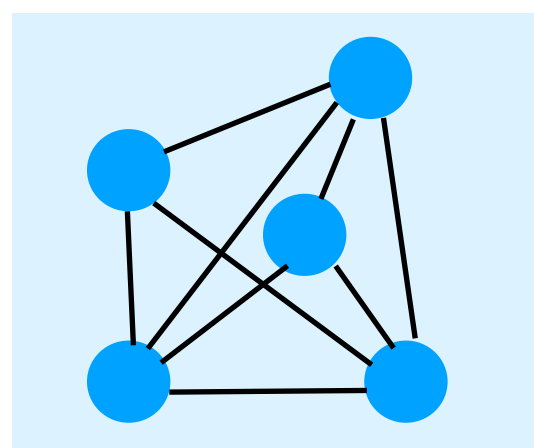
$$\alpha = V^T \cdot X$$

Sparse
Representation

	T1	T2	T3	T4	T5	T6
1						
2						
3						
4						

$N \times T$

Underlying
Graphical Relation



WE WISH
TO LEARN

Signals

	T1	T2	T3	T4	T5	T6
1						
2						
3						
4						

$N \times T$

$$X = V \cdot \alpha$$

Equation Relating both Worlds

$$\|X - V * \alpha\|_2^2 + \mu_1 \|\alpha\|_1$$

STEP 1

$$\min_{\alpha} \left\| X - V * \alpha \right\|_2^2 + \mu_1 \left\| \alpha \right\|_I$$

Solved Using CVX

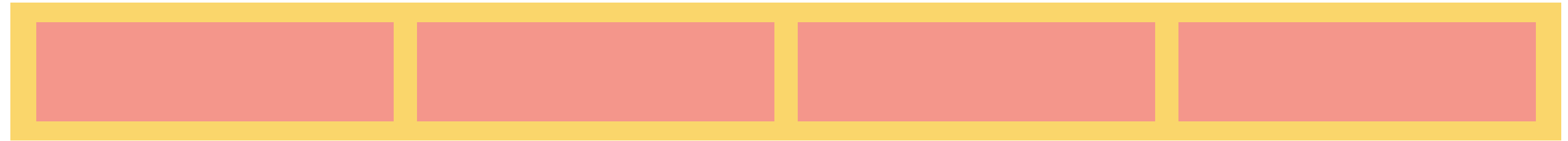
From this Step We Obtain

α

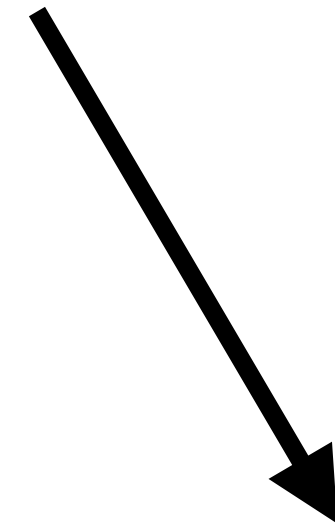
WE ARE NOT BANDLIMITED ANY MORE!

STEP 2

Notion Of Graph Sparsity and Connectivity

$\mathcal{L}$ 

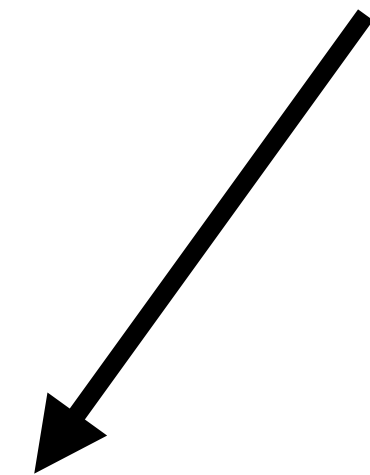
Element Wise L_1 norm
We Could Also Impose
 L_1/p Block Norm Regulariser



To Ensure Graph Stays Connected

$$-\mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T))$$

$$\|L\|_{1,1}$$



$$+ \mu_3 \|\mathbf{V} * \Delta * \mathbf{V}^T\|_{1,1}$$

But there is No **Data Fidelity** term, so to get in that we impose the following
 Here we try to Capture relation between **Eigen Values** and **Data**

$$+ \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * \mathbf{I} \right\|_F^2$$

Inverse Covariance Matrix

$$\hat{\theta} = \left(\frac{X \cdot X^T}{T-1} \right)^{-1}$$

Regulariser

$$\sigma = \min \text{EigenValue}(\hat{\theta})$$

*Formulation is being used in Literature

STEP 2

$$\min_{\Delta} -\mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T)) + \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{\text{I}, \text{I}} + \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * I \right\|_{\text{F}}^2$$

Subject To

$$1. \mathbf{V} * \Delta * \mathbf{V}^T * \mathbf{1} = \mathbf{0}$$

$$2. \Delta \geq 0$$

$$3. \text{Off_diag}(\mathbf{V} * \Delta * \mathbf{V}^T) \leq \mathbf{0}$$

Solved Using CVX

From this Step We Obtain

$$\Delta$$

STEP **3**

Putting up *Altogether* to Obtain the V

Use α and Δ obtained in previous step to obtain V

$$\begin{aligned} \min_{\mathbf{V}} \quad & \left\| \mathbf{x} - \mathbf{V} * \alpha \right\|_2^2 - \mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T)) \\ & + \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{\mathbf{I}, \mathbf{I}} + \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * \mathbf{I} \right\|_{\mathbf{F}}^2 \end{aligned}$$

Subject To

1. $\mathbf{V} * \Delta * \mathbf{V}^T * \mathbf{1} = \mathbf{0}$
2. $\mathbf{1}^T * \mathbf{V} = [1 \ 0 \ 0 \ 0 \ \dots \ 0]$
3. $\text{Off_diag}(\mathbf{V} * \Delta * \mathbf{V}^T) \leq \mathbf{0}$

We Solve this Using Manopt, with
ALM so as to impose Hard Constraints

Thus we compute the Laplacian from the learned V , Δ .

$$L = V * \Delta * V^T$$

$$W = \mathcal{F}(L)$$

$W_{\text{int}} \leq$ Make Diagonal of L all Zeros

W is Negative of W_{int}

Full Formulation At a Glance

We perform **triple-alternate-minimisation** over following **Cost function**

$$\min_{\alpha \Delta \mathbf{V}} \left\| \mathbf{X} - \mathbf{V} * \alpha \right\|_2^2 + \mu_1 \|\alpha\|_1 - \mu_2 \mathbf{1}^T * \log(\text{diag}(\mathbf{V} * \Delta * \mathbf{V}^T)) + \mu_3 \left\| \mathbf{V} * \Delta * \mathbf{V}^T \right\|_{1,1} + \mu_4 \left\| \hat{\theta} - \mathbf{V} * \Delta * \mathbf{V}^T - \frac{1}{\sigma^2} * I \right\|_F^2$$

Subject To

1. $\mathbf{V} * \Delta * \mathbf{V}^T * \mathbf{1} = \mathbf{0}$
2. $\mathbf{1}^T * \mathbf{V} = [1 \ 0 \ 0 \ 0 \dots 0]$
3. $\text{Off_diag}(\mathbf{V} * \Delta * \mathbf{V}^T) \leq \mathbf{0}$
4. $\Delta \geq 0$

Augmented Lagrangian Method

Consider solving the following constrained optimisation problem

$$\min f(\mathbf{x})$$

Subject To

$$c_i(\mathbf{x}) = 0 \quad \forall i \in \mathcal{E}, \quad \text{Where } \mathcal{E} \text{ denotes the indices for equality constraints}$$

This problem can be solved as a series of unconstrained minimisation problems. For reference, we first list the K^{th} step of the penalty method approach:

$$\min \Phi_k(\mathbf{x}) = f(\mathbf{x}) + \mu_k \sum_{i \in \mathcal{E}} c_i(\mathbf{x})^2$$

The penalty method solves this problem, then at the next iteration it re-solves the problem using a larger value of μ_k and using the old solution as the initial guess or "warm start".

The augmented Lagrangian method uses the following unconstrained objective:

$$\min \Phi_k(\mathbf{x}) = f(\mathbf{x}) + \frac{\mu_k}{2} \sum_{i \in \mathcal{E}} c_i(\mathbf{x})^2 + \sum_{i \in \mathcal{E}} \lambda_i c_i(\mathbf{x})$$

and after each iteration, in addition to updating μ_k , the variable λ is also updated according to the rule $\lambda_i \leftarrow \lambda_i + \mu_k c_i(\mathbf{x}_k)$ where $\mathbf{x}_k$ is the solution to the unconstrained problem at the k th step

